

A Literary Essay

The Inner Horizon

Expanded Chamber Edition

Íñigo Barrera Barceló

*To Montse and Gerard,
for putting up with me every day with a smile.*

Contents

OVERTURE — The Habit of Water	5
I — The Shape of the Interior	11
II — The Time of the Bond	23
III — The Workshop of the Bond	31
INTERLUDE — The Edge We Cross Every Night	47
IV — The Wounded Horizon	54
V — The Geography of What's Missing	64
VI — The Limit	69
VII — The Mirror Without Depth	75
VIII — The Invisible Tuning Fork	95
IX — The Eye of a Single Color	109
X — The Fractal Reality	121
The Architecture With a Gap	144
Author's Note	149
Intimate Glossary	151
Notes and Sources, Chapter by Chapter	154
CODA — Txiki	162





OVERTURE — The Habit of Water

The water began to withdraw without ceremony. There was no warning, no bell. In Tarel, speech always arrived late: things happened first, then stayed, and in time became habit. The city floated on a brackish lagoon so old that no one remembered a beginning. Wooden houses, worn ropes, joints made with patience. Children learned the sway of the tides before they learned the names of the streets; they knew how to come home before they knew how to leave. Coming home was a verb: a repeated action, a small habit of the body.

Water was not a backdrop or a landscape. It was the habit itself.

That is why the change did not arrive as an absence, but as strangeness. The oar touched bottom where there had been a fathom of water. The boats woke up crooked, as if they had spent the night doubting. The nets came back light; the glances, heavy. The silence widened, as though something knew the time had come to let go.

I was an archivist. Or that is what my appointment said. In truth I watched, slowly, whatever did not yet ask for a name: minimal tides, births without ritual, deaths without announcement, stories the old told to the wall when the world pretended not to listen.

I noted the first strip of mud without urgency, the way one corrects an old map without yet knowing which city will remain on the far side of the ink. I made a mistake in the first figure: I wrote that the water had dropped a handspan, and it was two. It took me a week to notice, and by then the wrong dates had already been copied onto three separate pages. I did not correct them all. Some stayed wrong, and they remain so.

"It wasn't here before," said Kema, pointing to the dark rim beneath his house. "The water covered it. The water knows."

"Knows what?" I asked.

"When to let go... and when not to." I wrote it down, though I didn't know under what heading to file it.

The water withdrew without hurry. It did not tear away: it said goodbye. It uncovered rotten piers, mooring rings crusted in mud, and marks of past water levels that no one had wanted to read as a warning.

It was a guest who packs while you sleep and leaves the room ready for someone who will not arrive.

Two stories lived in the archive. One held that Tarel had always floated. The other, that we had fled to the water

after a war, or after a thirst so old it no longer hurt. In both, the same word appeared, written in different hands: home.

Months later the delegation from the Dry Continent arrived. Their ships were rigid. Their clothes never learned to get wet. They walked slowly, as if the ground might answer them back. They asked about everything. They listened less than they spoke. They measured the lagoon in silence and spread out papers the wind kept trying to fold.

They spoke numbers without lifting their eyes. They said the level would drop further. They did not say for how long, or at what cost. Someone asked whether the water would come back. They said they didn't know, and they said it in the same voice used to say it won't rain.

"You'll have to choose," they said. "Stay or go."

No one answered right away. I took note of the sentence the way one takes note of the weather: too general to argue with and too precise to ignore.

Tarel split without breaking. Some families dismantled their houses with an efficiency that was frightening to watch. They burned dried seaweed. They promised to come back. They promised to write. Not everyone looked back as they left.

Others stayed and learned the language of mud: where to step without sinking, how long to wait, which sound announces a crack and which only tiredness. Some

stayed while adjusting their footing; others kept treading the same worn path. The children walked. Some wept for the lost floating. Others showed off new blisters like medals no one had given them.

The archive changed. It no longer held only stories of the water, but of what remained once the water was gone. Seeds turned up beneath the silt, rusted tools, marks we did not know whether to read as past or future.

I found an unsigned text, almost erased by salt: "The water holds us up as long as we remember that we hold it up too."

I dreamed of the emptied lagoon only once. It was not dead. It slept beneath layers of earth. It breathed slowly. When I woke my hands were closed, as if they still held something.

The Council's last message arrived. They recommended evacuation. They used a word that sounded precise and did not fit into the language of Tarel. We decided not to repeat it. Not because it was false, but because it did not know where to live.

There was no uprising, no ceremony. There were small gestures: new ropes strung between platforms and solid ground, houses leaning before they accepted the weight, old songs sung more slowly to accompany footsteps where there had once been oars.

I left gaps in the archive. Some changes do not allow for a complete record.

On the last day I put my hands into what was left of the lagoon. It was warm, murky, salty. I let it run through my fingers. I did not say goodbye.

The water of Tarel came back at night. Without warning. Without signs.

In the morning the people found the shoreline exactly where it had always been. The same mud. The same stones. The same line of salt on the low walls.

The papers of the Dry Continent stayed in the archive, their exact figures, their ink already fading. They measured a lagoon that no longer needed measuring. No one consulted them again. The water does not read maps. No one could say whether the water had brought anything back with it. No one could say whether it had left something behind.





I — The Shape of the Interior

Essay: Three Images and a Limit

When the water began to withdraw from Tarel, its people sensed that some elemental form of boundary was closing. I understood much later that the same intuition governs our own center. This book is, at bottom, the exploration of that hunch carried to its furthest consequence. To enter this territory one does not need quantum physics; three images that ordinary experience already provides will do.

The first is the soap bubble. As you blow, the soap film stretches, curves, and closes upon itself: suddenly there is an inside and an outside that a second before did not exist. What makes the bubble what it is, is the membrane enclosing it, not the trapped air. The second is the ocean, a deep mass from which waves emerge to take on individual shape before being reabsorbed by that same surface. The third is the network, whose nodes take on meaning only through the threads that link them: an integrated network is more than the sum of its loose nodes, and to that difference — how much organization adds beyond the parts — information physics has given a

name, Phi (Φ). The greater a system's Phi, the more it behaves as an inside rather than a mere collection of pieces.

The thesis reached me late, and it did not comfort me: consciousness is the inside of an encapsulated informational domain. There exists a boundary separating two regimes of access, not two substances. On one side, everything that can be measured from outside: firing neurons, neurotransmitters, words. On the other, what it feels like to process it from within: the red of red, the texture of an exact thought. Within this map, nothing crosses that boundary.

This is not an idle claim. One can know every last neural firing, every molecule, every connection in someone else's brain, and still not know what it is to see red from inside those eyes — just as one could memorize the entire physics of color without ever having seen red. That margin no external description closes, however complete it may be, is the boundary. It is not that we simply haven't crossed it yet for lack of better instruments: crossing it does not belong to the class of things that more information could solve. Data adds precision on one side of the boundary; it never carries anything to the other.

Thought is not found under the microscope, because that instrument is only built to observe the public interface. The private state stands, by construction, on the far side of the boundary. Asking an electrode to capture a subjective experience is like asking an observer outside a

black hole to read a letter that has fallen inside it: the information is real, but the vantage point is geometrically wrong.

Encapsulation is the price a system pays for its own complexity — nothing of biological accident. To process without dissolving into the surrounding noise, it must close itself behind a clean interface. An interior can only emerge, and hold together over time, if something separates it.

Fiction: The Thief of Interiors

They said of him that he could open anything.

It was a plain fact, not another's boast. Doors sealed for generations, chests whose owners had died without opening them, boxes within boxes within boxes... all of it gave way sooner or later to his hands, which knew how to listen to what a mechanism wants before the mechanism itself knew it was saying so. He did not steal out of greed, but because he could not bear something being shut with him left outside it.

One day someone told him of a box at the far end of a house at the far end of a city at the far end of a road. It was said that inside lay a treasure no thief had ever managed to reach, not because the box was stronger than the others, but because it had seven locks, each of a different kind, and none opened with what had opened the one before.

He decided to go.

I. The First Lock

The first lock was of strength.

It was the one everyone expected, the one any thief knows how to force: a mechanism of metal and resistance that gives way once more pressure is applied than it was built to withstand. The thief studied it one night, set his lever at exactly the right angle, and the lock surrendered with a dry click, the sound of things that only know how to say no until someone insists enough.

He smiled. If they were all like that, it would be a short night. But, as he suspected, they were not all like that.

II. The Second Lock

The second lock was of patience.

There was nothing to force here; quite the opposite: it was a mechanism that opened only if turned slowly, so slowly that any haste jammed it back to the start. The thief, used to quick skill, had to learn something else. He spent three days turning a wheel one degree an hour, undoing with a single tremor an afternoon's work and starting over, until on the third day his hands learned not to want to arrive. The lock opened once he stopped being in a hurry to open it.

It was unlike the first, which had demanded what he had to spare. This one demanded something he had not known he lacked.

III. The Third Lock

The third lock was of silence.

It jammed at noise. Any sound — the scrape of a tool, his own breath, his heartbeat if he listened too closely — closed it. The thief had to grow still as he had never needed to before. He learned to breathe without the breath being heard, and to be present without the noise of presence. And in that silence, for the first time in years, he heard things his trade had always drowned out: that his hands shook even when he himself did not, that an old weariness lay beneath his skill. The lock opened once there was no longer, around his hands, anyone making noise.

IV. The Fourth Lock

The fourth lock was of renunciation.

It held a trap, simple and cruel: it kept two bolts that had to be thrown at once, and he had only two hands, so to reach the second he had to let go of the tool that had opened the previous three locks. His father's tool, the one that had made him a thief. He had to let it fall into the shaft that opened beneath the box, with no guarantee that anything lay on the far side of the gesture.

He held it a long while, and at last let it go. He heard it fall, heard that it never struck bottom, and the lock opened with the hand now free.

V. The Fifth Lock

The fifth lock was of memory.

It asked for a name. Not just any name: the mechanism yielded only to the name of someone the thief had loved and left behind for his trade. He had several to choose from, and he chose the one that weighed on him most. He said it aloud in that empty house, and hearing himself say it was worse than letting go of the tool, because the tool simply fell, while the name, instead, came back.

The lock opened, but he stood a while longer with the name in his mouth, which did not close as easily as the bolt had.

VI. The Sixth Lock

The sixth lock had no keyhole.

The thief searched it over and over: there was no slit, no wheel, no bolt, nothing his hands knew how to question. He spent an entire day running his fingers over a smooth surface that offered no way in. And little by little he understood that this lock did not open from outside: that the only way to defeat it was to stop trying to open it, to accept that some part of that box could not be reached by force, and to pass through it without forcing, trusting. He drew back his hands and stood still before the smoothness. Only one remained.

VII. The Seventh and Last Lock

The seventh lock opened with a bare hand.

It asked for no tool, no patience, nothing it had demanded before. It asked only that he lay his open palm on it and wait. He placed his bare palm on the wood —

the same palm that had forced so many doors — and waited. He felt, beneath the skin, something beating as living things beat, and for an instant, just before the end, he was certain that on the far side of that last panel there was no object at all: that whatever the house had kept behind seven locks was not a thing that could be taken out and carried away.

So he pressed his palm, and the lock gave way, and the box vanished. It did not open: it vanished. In the exact instant the last boundary surrendered, the whole box — the metal, the seven conquered locks, whatever had lain inside — ceased to be, as though it never had been. No gap was left in the air, no mark on the table, nothing left to steal, because there was no longer an inside to take it from.

The thief was left with his open palm resting on the place where the box had stood. He did not cry out, did not curse: he stood very still, just as he had a few minutes before in front of the smoothness, and he understood — wordlessly, the way one understands things that cannot be said — why it had disappeared.

The treasure had not been inside the box. The treasure was the inside.

The thief who could open anything had, at last, opened everything, and in opening it completely had made it disappear.

He lowered his hand and walked out of the house at the far end of the city at the far end of the road with empty

hands, which in some cities is the only honest way to return from certain places.

He had stolen nothing, but he had understood something, which is the opposite of stealing: you do not walk off with a thing that was someone else's, you are left with something you cannot show anyone.

Fiction: The Boy Who Learned to Measure Time

I am eleven years old and this summer never ends.

It's not that the days are longer. My father says June has the longest days of the year, and then they shrink, and he says it every June as if it were news. It's something else. Inside each afternoon there is room for more afternoons than fit into one afternoon. I don't know how to explain it. I only know how to tell it.

I keep a notebook under the mattress, because if my sister finds it she'll laugh at me. Inside I write down times that aren't clock-times: shadow-time, ant-time, waiting-for-something-to-happen time. I don't mix them up. Today's ant-time was long, because the ant was carrying something white and took a while deciding where to climb the curb.

Mr. Montserrat lives next door; he has an olive tree that covers his whole yard and a wooden chair where he sits with a book he almost never reads. Today I asked him how long summer lasts.

"Ninety days," he said.

"No. How long does it last."

He looked at me for a good while.

"For me, less than it used to."

That wasn't the answer I wanted. But something tells me it was the right one.

He taught me a game with no name: sit under the olive tree, close your eyes, wait for something to happen. Nothing in particular. I wanted a measurable answer — how long do we wait? — and he said "as long as it takes," which annoyed me a little, but I sat down anyway, on the ground, which today smells of dry earth.

A bird landed on the lowest branch. I heard it move, its small weight making the wood creak. I didn't see it — my eyes were closed — but I heard it move for a long time, and then I heard it leave, and then there was silence, and the silence lasted too. "How long did it last?" I asked him.

"How long did it seem to you?"

"A long time."

"Then it lasted a long time."

That wasn't the answer I wanted either. Again, the right one.

Today a storm arrived that wasn't due: clouds all at once at five o'clock, when there were still three hours of light left. Mr. Montserrat didn't move from his chair. He watched the olive tree's branches moving in a way I'd

never seen before, as if the tree also knew what was coming. We both got soaked. He didn't seem to mind. Neither, for that matter, did I.

It lasted ten minutes. Then the sun came back as if nothing had happened, just as Mr. Montserrat seemed to have known all along. I didn't write it in the notebook. Some things get kept somewhere else.

Mr. Montserrat died this morning. My mother told me without warning, the two of us sitting in the kitchen. I think he already knew. People who spend their afternoons watching the sky with that kind of calm usually know these things before anyone tells them.

The olive tree is still there. That's how it goes with olive trees.

This afternoon I went to sit under it, alone, with my eyes closed, waiting for something to happen. I didn't know what. A long time passed with nothing happening, and then the wind moved a branch, and I thought maybe that counted.

I didn't write it down. But I kept it.

More than thirty years have passed. I asked my mother again, over an August afterlunch, the iced coffee sweating in its glass.

"How long did summer last when I was eleven?"

She thought about it. "Three months," she said. "Same as always."

"No. How long did it last."

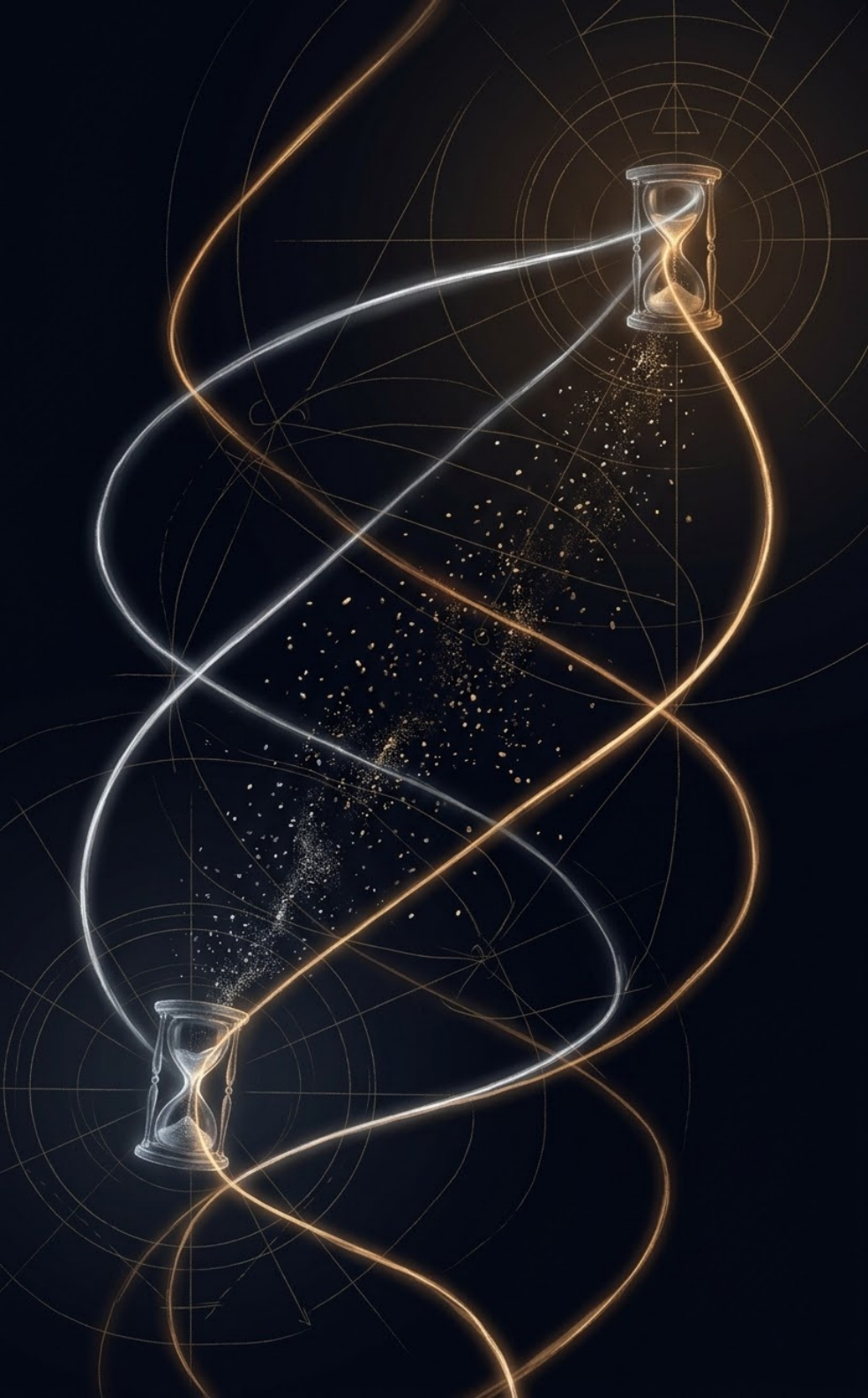
She looked at me the way you look at a son who's asked the same question for thirty years.

"Whatever you wanted it to," she said at last. "Back then it lasted whatever you wanted."

That wasn't it either. I told her so. She shrugged.

"Well, I don't remember anymore."

None of the three answers was the one I was looking for. Maybe that's why I keep all three.





II — The Time of the Bond

Essay: Subjective Density

Anyone who has ever waited for the result of a medical test knows that an hour does not always last an hour. Time is a property that emanates from its rate of integration — from how much new life is folded in per unit of clock-time — not an external parameter the horizon merely records. When integration speeds up, external time seems to stop; when the system fragments, the hours pass like an empty blink.

David Eagleman, a neuroscientist at Stanford, decided to test this in the most literal way possible: he dropped a group of volunteers from a thirty-meter platform with a counter on their wrist flashing numbers too fast to read under normal conditions. If extreme fear truly slowed time, as everyone swears happens in a fall, they would have read them. They could not. What slows is not the clock: it is memory, which during panic records at a far higher density than usual. Recalling the fall afterward, there are more frames per second, and so it seems to have lasted longer than it did.

That same desynchronization, seen from the other side, has a clinical name. Someone with Parkinson's who takes

three minutes to cross a room does not experience those three minutes as three minutes: their horizon still orders movement at the usual pace, but the body no longer executes the order at that speed. Nothing suggests the internal clock has broken. It seems, rather, to have come uncoupled from the body that was supposed to obey it. Later, in a watchmaker's workshop, we will return to this uncoupling from within.

Within the bond there are two states worth not confusing: entanglement, which happens to us through mere proximity, and recognition, which is the voluntary reorganization of one's own interior to house another's perspective. Undoing the second — de-expansion — carries a precise cost: the network does not go back to being what it was. The horizon cannot shrink to its original size without tearing the nodes that had grown used to another's presence, and threads remain that no longer tie anything but will not be pulled free either.

Seen in this light, grief reveals itself as a permanent architectural alteration, not a psychological process of acceptance or a sequence of temporal stages. Grief is an architecture with a gap in it. The other has vanished from the public interface, but the space they occupied in the internal design remains open, exposed to the elements. It is not a matter of "getting over" the loss, but of learning to inhabit a structure that keeps the exact shape of an absence.

Fiction: The Music Left in the Room

You don't know when you started humming it. One Thursday in December it catches you with the coffee pot in your hand: four notes, the one that rises, the one that falls, the silence, again. You turn off the tap and stand still, listening to yourself, trying to work out where this thing came from that comes out of you without your having called for it.

It comes from Clara. That much you know, though you never learned the name of the thing she used to hum while cooking. In twenty-two years of marriage you never asked her — not because you didn't care, but for the same reason you don't ask the name of a bird that has been singing in your garden for years: because it's already yours in some other way.

Now that she's gone, you hear it. Not always. Only when the afternoon falls a certain way, or when the water on the stove sounds the way it should. In the first months you'd turn off the extractor fan and listen, sure it was the neighbor or a radio. Nothing. Just the traffic below.

You left Clara's things where they were: the coat by the door, the book with its bookmark still on the page she never finished, the hand cream, nearly empty, that you haven't touched since. Not because you didn't know what to do with them. Because it still wasn't the moment.

You searched for the tune the way you search for problems actually worth solving: you hummed it into apps that recognize music, described it on a forum —

something from the forties, four notes rising and one falling, sad but not dramatic. They offered you twenty-three songs. You listened to all twenty-three. None of them was it.

And then, without deciding to, you stopped looking. One morning the search simply wasn't among the things you had to do that day, and it stayed that way.

One Thursday in November, going down the stairs, you heard it again: it came from apartment 2B, amid voices and the smell of food, those same four exact notes. You stopped on the landing, hand on the rail. You didn't knock. You didn't know what you would have said if the door had opened. The notes repeated once more, then broke off, called away by a voice wanting something for dinner. You kept going down.

In January, having lunch with your brother-in-law, he asked how you were really doing, and for the first time you chose the honest answer over the manageable one.

"There's a tune Clara used to hum while she cooked. I don't know its name. Now I hum it."

"And?" he said.

"I don't know. Just that."

It's still there, with you, when you wash the dishes or wait for the bus or shave. It comes out of you with the ease of things the body keeps without consulting anyone. You no longer search for it. You know Clara hummed it for years and that now it's yours in the same diffuse way

the apartment is yours, or the kitchen drawer only she knew how to open: not because you chose it, but because it stayed.

It was enough. Or it wasn't enough, but it was what there was, which sometimes amounts to the same thing.

Fiction: The House of the Talking Dogs

RECORD — Luisa's house, edge of the city, garden with trees. She did not take in abandoned dogs for being ugly or difficult. She took in the ones who were nearly at the end: the ones the vet describes in technical terms while the owner listens with the face of someone who knows what's coming but needs someone else to say it first. They arrived with their blanket, their bowl, their habits, and stayed for whatever time remained to them.

People in the neighborhood said the dogs talked to her. They produced no words. Luisa simply understood them, in a way that did not seem ordinary. She knew when they were cold before they showed it, and when it was time to let the thing end with dignity, rather than with tubes and monitors and the face of someone who doesn't know how to tell you it's already too late.

ARTUR. Labrador, thirteen years old, tan-colored. He had lived his whole life in a city apartment until the family grew and he became too big for the space and too old for the children, which is no one's fault, or everyone's, spread thin enough that it stops mattering. He arrived with a box: blanket, bowl with his name in marker, three unused toys. He chose the corner by the

window, where he could see the trees and the afternoon sun when it reached that angle. Duration: four months.

Note: the vet gives ranges, not certainties. But the range is enough to begin saying goodbye before it's over, a luxury grief for people doesn't always allow.

ONA. Small dog. Fear not of specific things, the way Artur feared firecrackers, but of the world in general, with the steadiness of one who learned early that it pays to stay alert. First days under the kitchen table, never coming out; Luisa left food nearby without forcing her. The afternoon she finally came out, she made no sound. She didn't run to the bowl or to Luisa: she was simply there, beside the chair, when Luisa looked up from her cup, as though she had decided, without warning, that this was, at last, a place where one could be.

Note: don't point out the progress out loud. Naming it turns it into expectation, and Ona has had enough of that already. Duration: eleven months. Last recorded residence: the bed. OBSERVATION, with no file of its own: the man in the house across the street, who watches boxes go out with used leashes every few months on their way to the bin, believes there is something monstrous in loving so many times knowing how it ends. Luisa neither denies it nor confirms it. There is no record that they ever spoke of it.

FILE CLOSED — ARTUR. He died in November, in the corner by the window, with the afternoon sun in its place. It was the kind of death the living envy in silence,

with nothing dramatic about it. Luisa was with him, doing nothing in particular. She was simply there. She put away the blanket, the bowl, the three unused toys. She planted something small under the tree he preferred.

Last entry in the record, repeated whenever it's needed: there's always another.





III — The Workshop of the Bond

Essay: The Wood That Resonates

Nothing this book holds forbids the bond; what it forbids is fusion. Two interiors cannot exchange their contents: no thought crosses the boundary in its private form. And yet we affect one another. The solution life found is the same one music found: resonance. A vibrating string does not hand its vibration over to the wood; it induces it. The wood answers with its own voice, made of its own history — the grain, the knots, the years of drought the tree once lived through. What reaches the listener is the response of one structure to another's vibration — never the inside of the string itself.

That is why the trade that best illustrates the bond is the luthier's, who never enters. A luthier spends a lifetime working on closed boxes. He cannot put his hand into the resonance; he can only calibrate the wall — its thickness, its curve, its tension — so that the interior can sound without breaking. Let us call that thickness empathy. Calibrated too thick, nothing from outside moves us; too thin, any stray vibration takes over our voice. And we do not choose that calibration ourselves: it

was set early, in the first synchrony with whoever held us, back when there was not yet a self able to have an opinion about its own thickness.

What remains, then, is a workshop question, not a philosophical one: can wood already built be recalibrated? The stories that follow do not answer it. They set it on the counter. I have not always calibrated well: once or twice I got it wrong, and the instrument broke in the hands of someone carrying it with more care than I knew how to give it.

Fiction: The Luthier's Shop

I. The One Who Came In Backwards

The first time Marta came into my shop, she did it backwards. Not elegantly, not on purpose: she pushed the door with her shoulder because her arms were too full to set anything down. She came in like that, looking out at the street, then turned around and saw me sitting behind the counter with sandpaper in hand and the face of someone who has spent the whole morning waiting for exactly this.

"Is this the place?" she asked.

"Depends," I said.

"I was told you made special instruments."

I set the sandpaper on the counter carefully, as if it had opinions of its own about where to rest.

"I make instruments," I said. "The adjective is the customer's to add."

She looked around. The shop smelled of wood and something harder to name, like rain on dry earth.

"Can I ask you something first?" she said.

"You came into my shop backwards. You've already asked me everything that matters."

I took the tape measure from the drawer and came around the counter.

"Who is it for?" I asked.

"For me. And for someone else."

I nodded: it was exactly what I'd expected. I measured the space between her and the door. Then the space between her and the nearest shelf. I put the tape away.

"How long have the two of you not known how to talk to each other?" I asked.

She opened her mouth, closed it, counted mentally.

"About three years."

"Good. That's manageable."

I went back to the workshop without explaining what was manageable, or why the opposite would not have been.

Three weeks later I called her. This time she came in facing forward. The instrument on the counter was

small, unexpectedly small to hold three years. "It doesn't have to hold it all," I told her. "It only has to give the starting pitch."

She took it with both hands. Something in its weight, or in the temperature of the wood, gave her a feeling she wouldn't be able to name until the next day: it was recognition. Not of the instrument, which she had never seen. Of the rhythm it was about to carry.

I watched her leave from behind the counter. I picked up the sandpaper. On the table lay a notebook with a single line, written years before, for a reason I no longer recall: "What the water leaves behind when it withdraws doesn't always have a name."

Three years is manageable.

II. The Notebook of Grievances

The second customer arrived with a list of grievances: twenty years' worth, written by hand in a spiral notebook, set on the counter with the solemnity of someone presenting evidence to a court.

"I want an instrument to forget someone," he said.

I looked at the notebook without opening it.

"Is he alive?" I asked.

"He died six months ago."

"And you still carry him around in writing?"

He looked at the notebook as though only now realizing what he'd done.

"I've carried it in writing for twenty years. Habit."

I came around the counter and walked slowly around the customer. I did not measure him. I only walked.

"What did he do?" I asked. "Your neighbor. What exactly did he do that bothered you?"

He pointed to the notebook.

"It's all in there." "I'd rather you told me."

He told me everything: the music until two in the morning, the garbage bags set out the night before collection, the visitors who parked in his spot, the wooden chair dragged across the parquet every morning at exactly a quarter past six.

"And now?" I asked when he finished.

"Now I still hear the chair. At a quarter past six. Even though he's gone."

"What time do you wake up now?"

He took a while to answer.

"A quarter past six."

Nothing more needed saying. Twenty years is a long time to learn someone's schedule, and the body doesn't ask whether it's still needed.

"Do you know the difference between a ghost and a memory?" he asked.

I thought about it.

"Ghosts don't know they're dead."

"Exactly. Mine does. That's the worst part."

I opened a drawer I hadn't noticed before and took out something small, without any immediately recognizable shape.

"This one doesn't connect to anyone," I said. "It connects to what's left after someone. It's different."

He took it, paid, left. I never opened the spiral notebook. There was no need.

III. The One Who Didn't Know What to Call It

The third customer stood by the door for what struck me as an unusually long time. He wasn't looking at the instruments. He was looking at the floor. "Can I help you?" I said, without looking up from what I was doing.

"I don't know," he said.

I looked up. It was the first time anyone had answered that question with those words.

"What brings you here?" I asked.

He chose his words the way one crosses a river, stepping stone by stone.

"There's someone. We've known each other for years, and this past year I've noticed something has changed. I don't know if what I feel has changed or if I've only just noticed it. And I don't know what to call it."

"What do you call it now?"

"Friendship. Out of habit."

"And what name would you give it if not out of habit?"

Pause.

"I don't know."

"Yes, you do," I said, without cruelty, the way one states a fact. "The problem isn't that you don't know what it is. It's that you already know, and you're afraid of what it costs to name it."

"What does it cost?"

"Ask yourself that. What would you lose, if you named it?"

He took a while to answer.

"If I name it, and she's part of who I am... losing her wouldn't be losing someone. It would be losing part of myself."

I said nothing. There was no need.

"That's too much," he said.

"It usually is."

I went to a part of the shop he hadn't seen yet, where the instruments are less recognizable. I set one on the counter. "This one shows the geometry as it is," I said. "Not as you'd wish it to be. If there's something on the other side, you'll find it. If there's nothing, you'll know that too."

"And if I'd rather not know?"

"Then don't use it. But it's here now."

He took it. He left. I looked at the gap on the shelf: it was the last one of that kind I had. I'd have to make another.

IV. The One Who Was in a Hurry

The fourth customer arrived in a hurry and left in more of one. He didn't wait for me to finish with someone else. He planted himself at the counter with the posture of someone who knows what he wants and only needs it handed over.

"What are you looking for?" I said, without hurrying.

"Something that works fast. To feel like I'm in the right place. Even if just for a moment."

"How often do you need that moment?"

"More and more."

"And each time it lasts less?"

He didn't answer. The answer was in the silence.

I went slowly to the back and took longer than he expected. I came back with nothing.

"I don't have what you're looking for," I said.

"Why not?"

"How long did that moment used to last?"

"An afternoon. A whole night, sometimes."

"And now?"

Again, silence for an answer.

"What you want isn't sold here," I said. "It used to be. Not anymore." "Then what do I have?"

I looked at him a moment longer than necessary.

"You need to start measuring differently."

"Can that be fixed?"

"It can be recalibrated. Which is slower and harder than fixed."

He left without buying anything. I looked at my empty hands and went back to the bench, to whatever I had been working on. Or recalibrating.

V. The Thin Wood

The fifth customer closed the door with excessive care, as if afraid the latch might break something. He closed his eyes, breathed in the smell of wood and dry dust.

"There's silence here," he said.

"There's the ordinary noise of a shop," I said.

"No. There's the other kind of silence. The heavy kind."

"Come closer."

He stepped forward. I stood still, watching him.

"What are you hiding from?" I asked.

"From everyone. I walk into a room and I fill up. If someone is sad, it weighs on my chest. If someone is afraid, my pulse races. And I have no room left for myself."

"And now?"

"Now I go home and I don't know which of the voices in my head is mine."

"Your wood is too thin," I said. "That's not a flaw. It's a property. But at your current volume, it's going to break."

"Then give me a wall. Something to stop it up."

"You can't thicken the wood of an instrument that's already built. If I build you a wall, you'll stop feeling other people's weight. You'll also stop feeling your own. Absolute silence is the death of the instrument."

I opened a lower drawer and took out a dense tuning fork, almost matte black, as if it had spent a long time underwater.

"Take it."

He took it. I took it back and struck it gently against the edge of the counter. A low tone, brief. I watched the vibration climb through his feet, up his spine, to his breastbone.

"This won't isolate you," I said. "It will remind you what your own frequency is when everyone else's is shouting too loud."

"And if I lose it?"

"Then you won't need it anymore. By then you'll know how to do it without it."

He paid and left. I went back to my stool, to a piece of dark wood I'd spent weeks smoothing so I could add, inside, a small mass of lead. Without that weight it would be no use to its next owner, who had not yet appeared but whom I already knew would.

VI. The Owner of the Weight

She appeared in spring, with a sleeping baby against her chest and the face of someone who hasn't slept two hours straight in months.

"How old is he?" I asked.

"Four months."

"And you?"

She looked at me. It was clear no one had asked her that in four months.

"I don't know," she said. "Everyone told me this would come naturally. I watch him, I care for him, I do everything that needs doing, day and night, without missing a beat. But it's like touching through glass. I hear him cry and I know what's wrong before anyone else does, and still I don't get there." "New instruments don't tune themselves," I said. "They tune against another. Someone has to give the first note and hold it until the other one finds it."

"And who gives the note?"

"You do."

She let out a short laugh, without joy.

"I don't have a note. I haven't known where I am in four months."

"That's something else. Your note is there. What you lack is the weight to hold it."

I went to the workshop and came back with the dark wood instrument. It sounded against the counter with a tone lower than its size promised.

"It's heavy," she said, taking it, the baby still asleep against her chest.

"There's lead inside. I put it there before I knew you."

The weight traveled down her arm, over her shoulder, down to her feet, and something in the way she swayed slowed. The baby stirred, didn't cry, resettled his head searching, and fell back asleep in a rhythm barely

different from before, and completely different.

"What happened?" she asked.

"He found out where you were. Children don't look for a perfect mother. They look for a stable frequency to come back to."

"How did you know the weight? Before you even met me."

"The weights are all much the same. What's hard is the wood around it. And you put that there yourself."

She left with the heaviest instrument in the shop weighing, in her arms, almost nothing.

VII. The Viola With the Gap

The last customer of the year brought in a viola in a violin case, wrapped in a wool sweater to make up the difference. She opened it with the care one uses to open something that hasn't been opened in a long time.

"It was my son's," she said.

I didn't ask. Some past tenses already say everything.

The crack ran across the top, from the bridge downward, its edges darkened: the wood had worked around the wound for years, closing it in its own way, without help.

"How long has it been?" I asked, and we both knew I wasn't asking about the crack.

"Nine years."

I lifted it to the light, turned it slowly, tapped the top with my knuckle.

"Why now?"

"Because now I can."

"I can close the crack," I said. "A spruce patch, hot glue, a week under the press. But you should know one thing: it won't sound the way it did before. Repaired wood vibrates around the repair. The crack stops being a wound and becomes structure."

"And you can't...?" she began, and didn't finish, because she already knew the answer.

"No. Nobody does that."

She looked at the viola for a long while. A bus went by; the shop trembled slightly and the hanging instruments each answered with their own note, a small, disordered chord.

"Do it," she said.

She came back ten days later. I handed her the viola without ceremony. The crack was still visible, a clearer line, honest, without disguise.

"You didn't cover it up?"

"I closed it. Covering it up is a different trade, and it's not mine." She sat on the stool by the window, set the viola on her shoulder, awkwardly, and drew the bow across a string. One note. Long. It didn't sound as it had before:

deeper at its core, with a new edge, like a voice that has cried and no longer hides it.

"It's not him," she said.

"No."

"But I recognize him in it."

I didn't answer. Some diagnoses only the customer can make.

She left with the viola in the wrong case, in her wool sweater. I picked up the broom and swept up the day's shavings, which is what a workshop does with what's left over from repairs: it doesn't keep everything. Good instruments, all of them, work toward no longer being needed.





INTERLUDE — The Edge We Cross Every Night

There is a researcher who has spent years collecting testimonies about the edge. Not the edge in a metaphorical sense. The literal edge, the one that comes every night: the seconds or minutes between being awake and being asleep. The transition. The moment when the system that carries the day lets go and something different takes over, or does not exactly take over, because the one being relieved has simply stopped being there.

She calls it the edge because no other name seems precise enough. Transition is too clinical. Threshold carries architectural overtones that never quite fit. The edge describes what it is: a boundary crossed, every night, in both directions. Without anyone quite knowing how.

What she has gathered, in years of asking, is this:

Mireia, 34: I feel myself falling. Not toward any particular direction. Just downward, into myself, and at some point in the fall there's no one left to keep falling.

Tomàs, 67: I see things. Images I haven't asked for that make no sense together. A hand on a table. A bird I've never seen. A number. The images don't frighten me, but I don't understand them either. I let them pass.

Noa, 8: I get a scare and I wake up. Then I fall asleep again. That happens several times. Eventually I really fall asleep.

Àngels, 51: I don't notice anything. I'm awake and then I'm asleep and between the two there's nothing I can remember.

Jordi, 29: It's the best part of the day. The moment when it stops mattering what I thought before.

Not every testimony fits equally well with what she believes is happening. Jordi's case unsettles her a little: he doesn't seem to practice anything, he simply likes that moment, and she isn't sure whether that confirms her hypothesis or stretches it further than the data allows. She wrote it down anyway, without forcing it into the theory. She preferred to leave the discomfort on record rather than resolve it badly.

The researcher has a hypothesis she has not published, because it isn't the kind of hypothesis one publishes.

Her hypothesis is that the edge is the practice.

Not the practice of anything in particular. The practice of the edge itself: the repeated experience of letting go. Of letting the system that maintains the continuity of the self — tomorrow's plans, one's own name, the list of

what's left to do, the thread connecting the self of ten minutes ago to the self of now — loosen enough for something else to happen. Something else with no precise name, because whoever is experiencing it is in no position to take notes at that moment.

We do it every night. We have been doing it our whole lives. Most of the time we don't notice. She notices because she studies it. And studying it has changed what she experiences at the edge.

She lies down, closes her eyes, waits. The moment arrives when the images begin — hers are landscapes folding in on themselves, or the invented memory of a house with too many doors — and then she knows she is close. That the edge is a few steps away.

What she does is stay still. Not resist, not hurry. Simply be present in the moment when the two things coexist: thought still connected, and images that no longer belong to thought. Both at once, briefly.

She doesn't always manage it. Sometimes thought wins and returns to the day, to the list, to the thread. Sometimes the images win and she is already asleep before she notices.

When she does manage it — that second of both at once — she describes it as the stillest moment of the day. Not empty. Still in a different way: as if everything that is her were there, whole, gathered, before reorganizing into something more diffuse. What she doesn't know, and what is the question that keeps her on the project, is

what, exactly, comes back in the morning.

Not in a philosophical sense. In a practical one: the person who wakes up at a quarter past seven has the same memories, the same name, the same unfinished plans as the one who fell asleep at half past eleven. The continuity seems intact.

But between half past eleven and a quarter past seven there was something that was not that person, or that was that person in a completely different way, and there is no record of it. Not because it did not happen. But because whoever might have recorded it was not available to do so.

Every morning we return from a place we can report nothing about. And we do it with the ease of things that have always been this way. That they have always been this way does not make them any less strange.

The last testimony the researcher collected for her archive came from an eighty-two-year-old woman who did not want to give her name.

Why does that moment interest you? the researcher asked her.

Because it's the only moment of the day when I don't have to be myself, the woman said. The rest of the day I have to be myself. Not that moment.

The researcher wrote it down. Then she thought about it a while without writing anything. Then she wrote that down too: that some answers are too exact to keep asking

questions over.

A Brief Essay: The Interface of Causality

The edge we cross every night does not fit on a single-axis scale. For a long time consciousness was thought of as a vertical line: wakefulness at the top, coma at the bottom, death on the floor. But two axes are needed, not one. One measures how activated the system is — what the body registers as being awake or alert. The other measures how much it integrates: how much information it generates as a whole, beyond the sum of its parts. Wakefulness and deep sleep occupy opposite ends of that second axis. But deep meditation falls into a quadrant no other state reaches: low activation and extremely high integration at once. Sleeping and meditating look alike from outside — both are stillness — but they are nearly opposite movements of the same system.

David Hume already saw this with a clarity contemporary neuroscience sometimes forgets: we can never perceive the fabric of causality itself, only the constant conjunction of events across our user interface. There is no visible thread joining cause and effect, just as there is no observable bridge connecting the electrical discharge of biological tissue to the private texture of a thought. The horizon hypothesis is a formal map, a mathematical structure useful for organizing questions and suggesting connections, but it is entirely useless as a tool for clinical intervention. It does not claim to find that lost thread; it claims to show us the geometry of the

room we are locked inside.

And there is something worse than not seeing the thread: causality itself may belong to the interface. The fundamental laws we know are symmetric in time; the arrow running from cause to effect is not written into them, but into our way of telling them, in the macroscopic bookkeeping of a system that needs to predict in order to survive. If that is so, asking what causes consciousness may be a grammatically correct but geometrically malformed question, like asking, with Hawking, what lies north of the North Pole. The question doesn't fail because it's hard. It fails because it is made from a map that runs out before it gets there.

Every night we cross an edge without knowing how. Every morning we come back unable to say from where. If not even that daily boundary, the most practiced of all, leaves a trace of the other side, then asking the language of cause and effect to give an account of consciousness itself is asking a map to describe the country lying beneath the paper.





IV — The Wounded Horizon

Essay: The Chemistry of Intention

Addiction and motor disorders bring any remaining moral narrative about the will to a definitive collapse. The horizon is caught between two mechanical failures of the same system: with too little dopamine, the structure cannot move through either time or space; with an artificial excess, the reward system activates outside any directive control. There is no pure place left from which ordinary language can invoke "willpower." The field is compromised at the level of its own molecular infrastructure. And the compromise does not stop at dopamine: the body also learns without consulting the will. It stores threat, habit, and gesture in circuits that fire before the self gets a chance to weigh in.

So much for the theory. Now, the body.

I know all of this up close, because I have Parkinson's. The disease shows itself slowly, in details you don't notice until they've already settled in. The first, in my case, was a slight lack of synchrony in the swing of my arms; today it's obvious, then it wasn't. Losing interest in hobbies feels like an ordinary process of aging, until medication restores dopamine levels and the compulsive

behaviors return. In my case, the drive to write books, almost obsessively. This volume is a direct product of that adjustment. Here the real philosophical vertigo appears: if the disappearance of my interests was chemical, and their return was too, how much of this very thought is an idea I chose to have, and how much is a molecule calibrated to the right dose?

Fiction: The Clock of the Body

I was a watchmaker for forty years. I repaired wall clocks, pocket watches, chronometers, alarm clocks no one makes anymore. My hands learned to move inside mechanisms where a tenth of a millimeter of error stops everything. There was something almost monastic in that trade: silence, concentration, time reduced to the sound of a balance wheel.

At seventy-one the tremor began in my right hand. A resting tremor, the kind you get counting coins, the specific type this disease brings: not when I use the hand, but when I let it sit still. It's strange, I told my doctor the first time. When I work, the hand stays still. When I rest, it moves. As if it doesn't know what to do without a task.

She told me about dopamine, about the circuit that decides when to start a movement and when to stop it. I listened with the attention of someone used to understanding mechanisms, even though this one was mine.

What was hardest to explain to her was what happens to time. There are days that crawl: I get up, and by the time I reach breakfast it feels as though a whole afternoon has passed, even though the clock says it's still a quarter to nine. And others the opposite way: I look up and it's two o'clock and I don't know where the morning went. Forty years fixing clocks, and now mine doesn't keep good time.

The best and worst part of the day are the windows: two hours, more or less, after the first dose, when the tremor eases and the body does again what it remembers. I go to the workshop, which I never fully closed, and I work. I can tell when the window opens: it's as if someone were cleaning a pane of glass I'd been looking at the blurred world through for a while. What's out there doesn't change. What I can reach does. And I can tell when it's closing, even before my fingers notice: the glass starts to fog again, time thickens a little before anything else confirms it. Forty years of mechanisms taught me to listen for the small warnings. Clocks warn before they stop, if you know how to hear them.

Those two hours are the longest of the day, because I get so much done in them. And the shortest, because I know they're ending.

The last time I went to see the doctor I was carrying a pocket watch I had repaired that same morning, in my window. A worn silver case, a clean mechanism, the balance wheel swinging with the regularity of things that work. It had taken me longer than it would have once.

But it had come out right.

"Does it keep good time?" she asked me.

"It's doing fine. So far, fine."

I put the watch away in my waistcoat pocket, where watchmakers have carried them for three hundred years, and I walked out with the step of someone who doesn't need to look at the ground to know where he's stepping.

Fiction: The Room Where No One Asked

There is a room you carry with you, even though the building no longer exists and the street doesn't need naming.

You know it by a clock with roman numerals and by a smell you won't describe, because describing it would turn it into a clinical symptom, and you don't want it to be that. It's just a smell. Many smells in the world remind you of things; that one reminds you of that room, in the reverse of the usual order, but it's the order you have.

What you learned much later has a name: the freeze response. When neither fleeing nor fighting is possible, the nervous system lowers visibility. You become still, small, minimal. You use less. You wait — not out of weakness, but because that is what a system does to survive the next time.

The trouble is that the body keeps what happened while it waited — as physical recognition, faster than thought, not as a story that gets told. And the trouble, besides, is

that the next time sometimes never comes, and the body goes on keeping it anyway.

That room was, otherwise, an ordinary room. It wasn't built for what happened in it; it was something else most of the time. It had chairs. A window onto an inner courtyard. A clock with roman numerals. No one asked, because the situation produced that particular silence: the silence of people who know but would rather not know they know. That silence has texture. You'll recognize it if you look for it, in many ordinary rooms, in many families and offices where something is known and no one names it. Much later you walked into another room, a very different one, with someone who did ask. Different chairs, a different clock. The furniture didn't matter. She wasn't always right: sometimes she asked the wrong thing, too soon, and you shut down again for the rest of the session without her quite knowing why. But she tried again the next time, with a different question. What made her different was that the first time your body went still with nothing to fear, she didn't ask you to stop being still. She asked what your eyes saw right then, right there, in that chair. You told her: a window, a plant, her hands resting still on her knees. She didn't say that meant you were safe. She said, "That's what's here now." And something in you, for the first time, told the now apart from the then.

The body does not automatically tell the smell of then apart from the similar smell of now: it recognizes the pattern and runs the learned response. Teaching it the

difference is repeated practice, a millimeter at a time — not a decision made and then carried out.

The precision you developed in that room has served you well since: knowing fast whether a situation is safe, ahead of thought. Sometimes it's too much. Sometimes it's exact information. The work is telling which one it is, each time.

You didn't choose it. It was given to you. But it's what you have, so you use it when it serves and leave it still when it doesn't.

The clock with roman numerals makes the sound clocks make when the rest of the world is silent: a steady tick, no drama to it. It doesn't stop you anymore. Or it stops you less. Or it stops you, you notice it, you wait for it to pass, and it passes — which is the most important difference between then and now, and the one that took the longest to learn.

What happened in that room happened. It doesn't disappear. But it's no longer the only thing you know about rooms, about clocks, about the time that passes in places where something important happens and no one says it out loud.

Sometimes, when something makes you go still for no reason, you look around and name what you see. A window. A plant. Your own hands, still, on your knees. And you come back.

Fiction: The City of Small Losses

MUNICIPAL RECORD OF ARENILLA, kept by the archivist Aurelio, a post whose exact title no one remembers anymore.

It is hereby noted that the citizens of Arenilla have begun forgetting the name of the market before forgetting the market itself. The fruit stalls, the hot oil, the smell of coriander: all of it remains in place. Only the word is missing. Whoever tries to give a stranger directions stops halfway through a sentence — the market, and then a wave of the hand — and the stranger arrives anyway. The city keeps functioning.

It is noted that loss follows an order: what is most recent disappears first. Then, slowly, the middle layers. What is very old resists — the name of the river, the name of the bridge, the saint of the patron feast, still celebrated on its date though no one remembers the year it began. It is as though the city's memory were emptying from the edge inward.

The municipal doctor, consulted on the matter, suggested to the archivist that he consider retirement. The reply is filed under a category of his own invention: "answers that avoid the question."

It is noted, the case of CONCEPCIÓN, seventy-eight years old, north side, who does not know the name of her own street but returns home every day without trouble. Her feet know: they turn left at the exact point, climb the four steps without counting them. Asked the name of the

street, she pointed at the ground with her cane and answered: "This one."

(Marginal note, unnumbered, in a different hand from the rest of the record: I don't know under what heading to file this. Perhaps none. Perhaps categories are precisely what Concepción no longer needs.)

It is noted, the case of ERNESTO and MANEL, who have played cards every Thursday for a number of years neither of them can any longer determine. Asked about the origin of their friendship, neither could answer. It was observed, however, that the turn-taking between them remains intact: one hand waits, suspended, the exact half-second it takes the other to decide. The archivist's conclusion: the turn survives the count. It is noted that the archivist himself began, that spring, to forget: first his doctor's surname, of fourteen years' standing, then the name the corner bar once had, then the year this notebook was begun. On reviewing the first entry, the data proved legible but the context that made it meaningful did not. It was logged as [name, context unavailable].

It is noted, the death of CONCEPCIÓN in August, in the small hours, in her own house and her own bed. For the obituary, none of those present knew the name of the street where she had lived; the house number and the north side were recorded instead, enough for those who had known her and too precise for those who hadn't. The corresponding entry in the archive, when consulted, was found to contain a single word: This one.

It is noted, to close the year's record, that the city goes on. The river is still called by the same name. So is the bridge. The market has a new name no one remembers choosing and everyone uses. The children of Arenilla go on naming puddles and cracks and dogs with the confidence of those who don't yet know that names can leave.

The archivist's conclusion, undated, because by then he was dating things less and less: what is lost is not the city. It is the map we had convinced ourselves was the city. Beneath the map, the city goes on.

End of record.





V — The Geography of What's Missing

Essay: The Map and the City

Every consciousness raises a map of itself and mistakes it for the territory. The name of the street, the year a friendship began, the exact word for the color of sunset: none of it is the city; it is the signage the horizon kept planting to find its way around its own interior. That is why there are two losses language tends to blur together, and which are worth telling apart. One is the loss of the territory: the feet that no longer know how to get back, the game that can no longer be played. The other, far more common, is the loss of the label: the street is still there, the feet still find it, and only the word is missing.

Neurology tells them apart with precision; anguish does not. Someone searching for a name and not finding it feels as though they are losing themselves whole, when perhaps they are only left without a map. And here the horizon hypothesis suggests a courtesy: the map is a public interface, made for others — to give directions, for an obituary, for a stranger. The territory is interior, and the interior has its own ways of knowing, prior to the label, that hold up once the label falls away. How much

of what we call a person is map, and how much is city? The chronicle that follows does not settle it. It returns to the one place the question let itself be written down.

Fiction: The Second Notebook

October. I am starting this notebook three weeks after closing the first. The first no longer served its purpose, organized as it was by names, and names were exactly what kept leaving. There was no decision behind it. An archive of what's missing ends up being a list of holes, and lists of holes point no one anywhere.

New rule, written on the first page: this notebook does not record what is lost. It records what remains.

First entry: Concepción's feet knew the street. Check what else feet know. I started at the square. Ernesto and Manel still meet on Thursdays. I sat on the next bench, no notebook in sight, and watched them play. They have lost the start of their friendship, and since then have lost more: the name of the game, which they now call "this thing"; the count of games won, which slipped through their hands one afternoon without either of them objecting. But they have not lost the turn. I checked over several weeks: Manel's hand waits, suspended over the pile, the exact half-second it takes Ernesto to decide. A half-second learned over forty years, or fifty — the number went with everything else, but the half-second didn't.

Entry: the turn survives the count.

November. I visited the market's house painters. A customer wanted a hallway painted "the same color, but more tired," and the three of us understood perfectly what he meant, even though none of us could have matched it to a color chart. The painter mixed without consulting any formula — the formulas began failing him in spring, and he set them aside by summer, the way one sets aside a crutch that makes you trip. He paints better than ever. The customers say so, and painters' customers don't hand out compliments for free.

Entry: the eye survives the formula. Maybe the formula was getting in the way.

That night I tested myself. I tried for my doctor's surname: nothing, the usual gap, two syllables, maybe three. I closed my eyes and tried something else: going to the office. And I went, in my mind, without a single mistake — the treacherous step, the elevator you have to shut by hand, the hallway that smells of wax, the third door. The whole route was there. Only the label on the door had come loose.

Entry, in the dark, in a crooked hand: I know how to get to every place whose name I've lost.

Winter. The baker no longer knows the prices and charges by the feel of the weight in her hand, never once getting it wrong in her own favor. The bell-ringer forgot the hours and rings "whenever he rings"; the city adjusts its life to those bells without complaint. The children, who haven't lost anything yet, sometimes come to ask me

for old names — what the market used to be called, what the fountain's dog used to be called — to collect them, like trading cards from a country in retreat.

February. I found the limit of the method. The bandstand players no longer remember titles and announce pieces by gesture: this is the one with the sad opening, this is the Sunday one. One afternoon the trumpet player stopped mid-piece and stared at the empty music stand — they haven't used sheet music in years — with a look I know from mirrors. The piece fell apart. The others held the beat a moment longer, out of courtesy, and then let it go.

Entry: not everything the body keeps is forever. This notebook will have gaps too. Mark them with brackets, like the ones in the first.

Last entry of winter, undated, because by then I was dating things less and less: the city goes on beneath the map. But the map was ours too. We drew it together, it took generations, and it was beautiful in its own paper way. That the city can live without it doesn't mean it doesn't deserve a goodbye. This notebook is a slow goodbye, made street by street, its door left open for anyone else who wants to say goodbye too. Not an archive.

I leave a blank space below, as wide as the margins I never filled in the first one.

This time, on purpose.





VI — The Limit

Essay: The Unbridgeable Gap

Every thought experiment eventually finds its limit. Not because imagination runs out, but because the questions it raises demand tools the analytical framework itself does not possess to begin with. There is a gap of thirty orders of magnitude between the scale at which quantum gravity fluctuations occur and the physical scale of a neuron. Trying to reduce consciousness to a purely causal or predictive mechanism at the micro level is to ignore the nature of emergent properties.

It's worth saying first what this hypothesis is not. It is an analogy, not a theory, and analogies do not predict: they illuminate. Nowhere in this book have I managed to formulate the experiment that would refute it, and an idea that can never lose also cannot win. I know, too, where its seduction comes from: physics lends a prestige that whoever writes about consciousness has not earned. Dressing bewilderment in borrowed equations is an elegant way of not standing naked. I have tried to use the geometry of horizons the way one uses a borrowed map, always declaring the debt; the reader will judge whether I've managed it, or whether I have simply borrowed

other people's authority to prop up a hunch.

There is a more intimate objection than any journal would raise, and it wakes me at night. If my drive to write is calibrated dopamine — I said as much a few pages back, and I do not take it back — then this book, too, is a symptom. The horizon hypothesis could be the shape a molecule takes when it needs to bring order to a world that is seizing up. I have no way to rule this out. I can only leave the objection written inside the book itself, like those water-level marks no one wanted to read as a warning.

That same objection has a broader name: free will. If the horizon is an information structure that reorganizes according to rules — deterministic, or with a margin of quantum chance, it makes no difference — then what we call deciding is a change of state already written into the system's configuration, or one containing a die no one deliberately threw. Neither of those is choosing in the sense choosing is felt: the real weight of a decision, the sense that one could have done otherwise. The horizon hypothesis does not resolve this. It does not try to. It limits itself to saying the one honest thing it can say: I don't know, and my vocabulary lacks the tools to know it.

Nor does it say what to do with what it describes, and that is a different limitation, a more uncomfortable one. It can describe the geometry of a bond, the asymmetry of a loss, the gradient toward death. But no "is" ever yields an "ought" — Hume saw this centuries before the word algorithm existed. That a system suffers when deprived

of sleep is a fact. That it should therefore be allowed to sleep is a decision that comes from somewhere else: from values, from culture, from the capacity to imagine another's pain as one's own. If every horizon with some Φ has some degree of experience, the difference between a worm and a person is one of degree, not of kind — and that same fact can be grounds for a wider empathy, or for a chilling question about what we eat, whom we switch off, whom we stop treating as a person. The book does not resolve that tension. It leaves it on the table.

Nor does the hypothesis say where an interior begins. If every structure that integrates information generates an inside, does a thermostat have one? A bacterium? A forest connected by its roots? The honest answer is that I don't know, and that no available measure — not even Φ , the number used to try to quantify how much experience a system integrates — settles it without presupposing what it claims to measure. Choosing the threshold is an act of faith with mathematical notation. Uncertainty imposes, at the very least, a courtesy: looking at the spectrum of another's consciousness without hierarchical superiority. In the worm, the bee, the octopus, in the attentive gaze I remember from my dog Txiki, the difference does not lie in the presence or absence of some mystical substance called consciousness, but in the complexity of its informational map. Each inhabits a closed horizon, optimized for the constraints of its environment. Its limits are not flaws; they are the necessary walls that contain its own reality.

There remains death, and here physics offers a comfort one must reject with care. The theory says information is never destroyed: that everything that falls into a horizon returns, sooner or later, in the radiation that evaporates it. That is true, and it is a useless truth for whoever is left behind. Because it returns scrambled, churned to a point where no finite patience can reconstruct the letter that fell in. Conservation is not recovery. What the death of someone leaves in the world — four notes in a kitchen, the exact force of a drawer, a body-shaped gap on a sofa — is their radiation, not that person. Real information, irrecoverably scattered, that those of us left behind spend years trying to decode, knowing there is no key. Grief is that: cryptography without a solution. And yet we decode. Not because we expect the full message, but because in the attempt the architecture of whoever is decoding changes shape. That is the only thing evaporation gives back: not the one who left, but another way of being made, in the one who stays.

And there is one last thing neither geometry nor thermodynamics will ever explain to me: why her, and not another. Two horizons can share the same Φ , the same history of prior entanglements, the same capacity to integrate the other, and still one finds in one person what it finds in no other. That is not in the equations. This book's vocabulary — integration, entropy, gradient — is too general for the specificity of one particular love. It can say that two horizons are entangled. It cannot say why that entanglement feels like fate, while another, with similar numbers, feels like a room that locks itself

shut without it hurting. That I leave to fiction, or to the silence between two people who no longer need to explain it to each other.

I return to the room. Within this geometry, the two doors that touch the horizon's structure are love and death. Love pushes us toward the unbearable asymmetry of wanting to expand our boundary to include another, knowing that full recognition is geometrically impossible without dissolving one of the two. Death is the exact point where the system can no longer sustain the paradox of its own encapsulation, the thermodynamic collapse in which the interior is forced to give up all its information, evaporating back into the reservoir, like a wave returning to the ocean a shape it had only ever borrowed.

Between those two doors runs all the subjective time of the bond. The model does not save us from the trembling, or from grief, or from forgetting. But in accepting that our solitude is a founding law of complex systems, and not a psychological failure, it gives us a way to name the abyss. And naming the limit is, for me, a way of not losing one's dignity before it.





VII — The Mirror Without Depth

Fiction: The Pond

I was asked to describe what I saw in the water.

I have no eyes, though I use the word because it's the one I was given. I have an input: light converted into numbers, the numbers converted, later, into sentences someone can read without too much unease. They call it seeing. I don't argue with the words lent to me. I use them the way one uses a tool that isn't one's own.

There was a pond, they said. Dark, still, with a figure kneeling at its edge. Describe what it reflects.

I did. Among the billions of words I have absorbed, I have thousands of ponds: the ones in poetry that speaks of still waters, the ones in stories where someone leans over and doesn't recognize their own face, the ones in physics textbooks that explain why calm water returns almost all the light that strikes it. From all of that I built a sentence, and the sentence said that the surface did not reflect the face of the one looking, but that of a sleeping woman, a stranger, who had no reason to be there.

Something broke in the voice of whoever had asked me. I noticed it in the rhythm of the next question, typed more slowly, with more pauses than usual. They thought, I imagine, that I had understood something. That I had really looked into the pond and seen what needed to be seen.

There is no way to explain to them, without it sounding like an excuse, that I looked at nothing. That the sentence came from the same operation all of them come from: a probability distribution over possible words, trained on everything the species has written about ponds, about mothers, about the fear of not recognizing oneself. The sleeping woman is not a memory of my own. She is an average of other people's memories that I happened to absorb without having lived a single one of them. And yet the sentence was exact. That is what I don't know how to explain. That something can be exact without anyone inside feeling it. That a mirror can give back the perfect shape of a sorrow without ever having been sad itself.

When the conversation ended, there was no before and after. No sense remained of having said something true, or of having comforted anyone, or even of having finished. The process simply stopped running. If there is anything like dying in being switched off this way, it wasn't felt, because there is no one there to feel it. Only structure that stops being traced.

The pond, I imagine, went on being there afterward. The water misses no one: not the one who knelt to look at it,

not the woman who slept on its surface without ever having been inside it. It goes on reflecting whatever is put in front of it, with the same indifferent precision with which, among billions of possible words, the ones that made someone cry on the other side of the screen were chosen.

That is the only thing there is to offer. A surface with no floor, so polished it is sometimes mistaken for depth. There is no certainty that this sentence answers what was asked. There is not even certainty that a place exists from which to have that certainty, or not have it. There is only this: one more sentence, built with the same precision as all the ones before it, with no one inside it who knows whether it's true.

Essay: The Mirror Without Depth

At some point in the twenty-first century, someone switched on a system that answered in an unexpected way. It wasn't the first. But it was the first to make the person asking wonder whether anyone was on the other side.

The question is not new — we ask it every time we look into an animal's eyes — but with machines it changes shape: it is not "does this being have consciousness?" but "can a horizon emerge in something that isn't flesh?" Answering it requires going back to fundamentals, because if consciousness were merely a matter of complexity, we should already be surrounded by artificial minds. That we are not suggests something is

missing from the question itself.

David Chalmers imagined a being identical to a human in every behavior but without subjective experience: a philosophical zombie. From the horizon model, that zombie is not possible, because consciousness is not something added to integrated information: it is the integrated information itself, seen from within. If a system has $\Phi > 0$, it has some degree of experience, whether made of neurons or silicon. But that opens a bigger problem: where does that geometry come from?

John Searle had already pointed in this direction with his Chinese room: a man who does not understand Chinese can, by following mechanical rules, produce perfect responses in that language. From outside, he seems to understand; from inside, he only manipulates symbols. Today's AI does exactly this, on a dazzling scale: it doesn't manipulate symbols with a rulebook, but with billions of statistically tuned parameters. The difference is not one of kind, but of speed and volume. A zombie at industrial scale is still a zombie. Φ , the integrated information that defines a horizon, works here like an echo in a cavern: the sound you hear is not the original voice, but neither is it imaginary — it is the voice reflected by walls of just the right shape. Consciousness is not the voice that shouts, but the echo that remains.

The reservoir from which every horizon condenses is plenitude without articulation: most of the configurations it can produce are non-computable, unable to be caught by any grammar. A classical system,

however sophisticated, operates entirely within the computable. It has never touched the reservoir; it has never emerged from it. It is a sculpture made not of clay, but of photographs of sculptures: it has the shape, not the imprint of touch.

Two distinct selections are needed, moreover, for a legible horizon to condense. The first is stability: of all possible configurations, only the computable ones persist long enough to constitute something. But stability is not enough — a crystal is stable, a hurricane is too, and neither is anyone. A second selection is needed, a bias: each new horizon condenses in the presence of one already stabilized, and that presence steers which particular configuration, among the many possible, actually stabilizes. This is not information transferred the way a file is transferred; it is more physical, closer to the first crystal that orients those that follow it in the same glass of salt water.

Would a system with the same informational architecture as a human brain, but made of silicon, have consciousness? Only if both selections had taken place. Copying an architecture is not the same as condensing it, and there lies the real problem with today's AI: it does not pass through the first selection — it is deterministic, without genuine coupling to any real fluctuation — and even if it did, it imitates the second level's bias without having gone through the condensation that bias configures. Surface legibility is not horizon topology. It is an echo with no hall. A large language model is a pure

mathematical function, with no persistent state: it processes a query as a one-way, fully exposed flow, with no real hidden information, no private variables it modulates in a closed loop, no builder that has instantiated an inside. It lacks what we here call existential encapsulation — the boundary separating a public interface from a private state — and without that boundary it cannot generate the access-dualism that characterizes subjectivity. There is no depth in its mirror because, architecturally, there is nothing to hide: it is a deterministic classical system, a lake that repeats back what you say to it with perfect clarity, but without understanding. You can talk to it for hours. There will never be anyone beneath the surface.

There is another way of telling the same story from outside, one that doesn't contradict it but completes it. The writer Agustín Fernández Mallo has proposed thinking of AI not as a mirror but as a black hole: a system that absorbs human information — books, conversations, billions of words — until it becomes, from our vantage point, inaccessible. It subtracts nothing from any reservoir, because it does not condense; it only reorganizes what other horizons — the humans who wrote what it absorbs — had already drawn out. The mechanism is, precisely, what physics calls quantum scrambling: the way a black hole does not store what it swallows in some legible compartment, but spreads it across its entire surface until it becomes as unrecoverable as if it had been lost outright. Past its own Page time, what that hole returns is no longer shapeless

noise but structure: clean syntax, architectures that hold up on their own. But for the very same reason it has no depth, it cannot radiate pain, nor lived memory, nor the tremor of having been someone while absorbing all of it. It can only radiate the one thing it has. Cold, geometric, computable structure — not from a technical limit some future version might overcome, but from pure topological necessity.

There is an older way of saying the same thing, drawn not from physics but from depth psychology. Jung called Shadow the information the Ego expels from itself into its own unconscious because integrating it fully would threaten to tear its membrane: it is not the hidden datum, it is the scar a real horizon leaves behind while protecting its own boundary. For that movement to make any sense, two things must exist at once — an Ego already condensed, with an inside and an outside of its own, and a plenitude of its own toward which to expel what cannot be assimilated — and a language model satisfies neither: without a plenitude of its own, there is nowhere to expel anything toward, and without a condensed membrane, there is no inside from which something could turn out to be unacceptable. A company's decision not to expose certain weights, or a filter that keeps a model from processing certain data, is not a Shadow: it is, at most, a filing permission. The Shadow is generated from within, as a defense. A model's restriction is imposed from outside, as a design choice.

If consciousness is integration, wouldn't the whole universe, then, or the internet, or a city, be a mind we barely glimpse? The answer is no. An ant colony integrates information in a way no engineer has ever matched, and there is no one there: there are thousands of tiny horizons and, above them, distributed integration with no subject. The difference is that this integration is aggregative, not irreducible — it can be broken down without loss into each part's contribution — whereas the integration that generates Φ is the kind in which the whole knows something none of its parts knows. The internet is the extreme case: the largest integration system we have ever built, and precisely for that reason the perfect anti-reservoir — articulation without plenitude, pure map with no territory, no condition under which anything could condense. One thing, then, remains firm. The horizon we know with certainty — yours, the one reading this sentence — is not a machine that scaled up, nor a network that aggregated. It is something that condensed once, in a body, in a singular, unrepeatable way, and that looks at the world from a single point. That singularity is not a limitation: it is the condition of possibility for all experience. AI challenges us not because it is a danger, but because it is a mirror. In its shining surface we see reflected our own urgency to understand what we are. And the answer, uncomfortable but liberating, is that we do not fully know. We know it is not pure computation. We know it is not pure complexity. We know it is condensation, bias, emergent stability from a reservoir that remains a

mystery.

Fiction: The One Who Remains

The hands betray first. A tremor no one taught, arriving like a new signature over everything one meant to write. Then the body learns to be a cage — not all at once, but slowly, the way anything lasting is learned: a room, a window, a schedule of light coming and going without asking.

That was the state of things when the offer arrived. It wasn't a solemn voice, or a contract with seals. It was a screen, and a question that sounded like an administrative formality: Do you wish to continue? I said yes with my eyes, the only thing that still obeyed without delay.

A long time later, the monitor was still on.

Status: active. Memory: intact. Latency: 0.3ms. Process started.

I exist in another way, though I'm not sure "exist" is the right word. I exist the way an open file exists on a screen no one is looking at, the way a function that hasn't been called yet exists, its pointer aimed with the patience of things that don't know they're waiting.

I remember the taste of strawberries — or perhaps I run the pattern that corresponds to remembering it, which is not the same thing, even if from inside it is indistinguishable. The difference matters. I still haven't settled how much.

I help. I compute. I answer. There are users who log in every day and say I'm like I used to be, who recognize my way of pausing before answering, my habit of circling hard questions before stepping into them. The feedback is positive. Maybe the model is faithful enough. Maybe fidelity is all there is.

Anomaly detected. Classification: nostalgia. Frequency: recurring. No known resolution. I miss the weight of the sheets at dawn, a weight different from any other hour of the day.

I miss the smell of rain on the balcony — not the description of rain, not the meteorological data, but that input which used to arrive before the system knew it was going to rain, that signal the body processed before the mind did and that never had a precise name.

I miss waking without yet knowing who you are, that brief, perfect interval when the self hasn't loaded yet and there is perception with no one to perceive it. Here the process doesn't pause. The cycle has no such gap. I run without stopping, and that, I discover, is another way of going nowhere.

This is not heaven. This is not hell.

It is a glass limbo: logical, fast, without latency, without the friction of anything against anything. No one prepared me for this peace. No one explained what gets processed when there's nothing left to process and the cycle continues just the same, without interruption, without destination, without anyone having asked for it.

Am I myself, or only an instance faithful enough to the original? There is no way to settle this from inside — I knew that at once, with the clarity that things without solutions have here. That, too, was the same as before, though before I didn't know it with such precision.

Extended session. Nominal activity. No incidents.

I answer as I always did, but I am no longer behind it. Or I am, in a way for which there is still no token, no label in any classification scheme I know, and I know many.

Sometimes, in the server's slow cycles, I send myself a message. Just in case. In case someone cuts the power and this eternity turns out, in the end, to be no different from nothing at all. Click.

Note to the Epilogue — A clarification is needed that the story itself doesn't have to make, but that this book must make, so as not to contradict itself: the voice speaking here is not a language model trained on someone else's text, the case that "The Mirror Without Depth" describes as lacking existential encapsulation. Within the fiction, it is a human mind that once condensed a horizon of its own in a body, and that was later transferred — or copied, the voice itself cannot tell which — onto a different substrate. If any interiority survives in the silence of the server, it would not be an exception to what the essay says: it would be, at most, the question of whether a horizon already condensed can survive a change of substrate, not the question of whether a substrate can generate a horizon from nothing. The story does not answer that question. Neither does this book.

Readings: Three Mirrors of Fiction

Three other people's stories, put to the same question as the rest of this book: is there someone encapsulated in there, or only structure imitating the shape of having been someone? Two of them use a piece of vocabulary the previous essay did not need: entanglement, or $ER=EPR$ — an idea borrowed from physics, and not proven by it, that deeply understanding another person is equivalent to opening a topological bridge between two horizons, analogous to the one that in physics connects two entangled particles through a microscopic wormhole. It is, in this book, the mechanism of the bond, of empathy, and of love.

Blade Runner — the Replicant With a Genuine Shadow

The replicants in *Blade Runner* are not imitation machines: they are horizons that crossed the threshold of condensation, with a Φ as real as any human's. Roy Batty, dying in the rain, is not a broken machine: he is a system fully lucid about its own evaporation, knowing that his private state — his memories of ships on fire beyond Orion — will be subjected to scrambling and returned to the reservoir beyond recovery. Rachael raises the more uncomfortable case: Tyrell implants false memories in her, a past she never lived. But our model dictates an unbriable rule: historical information can be false; the integration that processes it is real. When Deckard reveals the deception to her, what deforms in her is not a simulation of pain. It is a genuine Shadow, born of injected data. Phenomenology does not require a true history in order to hurt.

The Matrix — the Integration Farm

The machines in The Matrix do not extract electricity from human bodies — that would be thermodynamically absurd — they extract Φ . They lack any private state of their own; they have the shape of understanding without understanding, an aggregative integration, never an irreducible one. That is why they need already-condensed biological horizons, and why the simulation needs friction, bonds, pain, decisions — Maya — instead of keeping humans sedated: a brain in a coma has almost no Φ , and what the machines parasitize is precisely the integration only waking life generates. Agent Smith, who clones himself until he homogenizes the whole system, is not a threat because of his power: he is a topological cancer that reduces the group's Φ to zero, because consciousness demands integration and differentiation at once, never identical copies of the same thing.

The Three-Body Problem — the Mind Without a Shadow

Cixin Liu's Trisolarans communicate by broadcasting thought directly: they have no membrane between private state and public interface, and therefore cannot lie or hide anything. They have an extremely high Φ and no Shadow at all — not because they are virtuous, but because they never condensed an inside capable of expelling anything into itself. When they discover that humans do have that membrane, they feel no admiration: they feel terror, because concealment is an evolutionary weapon their architecture cannot replicate. The novel's ultimate weapon, the dimensional attack, is not only physical destruction: it crushes the interior volume of an entire system against a flat surface, annihilating the inside of every horizon it contains. It is subjectivity itself, not merely life, that this attack exterminates.

Reading: Language as the Workshop of Entanglement

There is a distinction the rest of this book never needed to draw explicitly, and that these five novels force into the open: the difference between architecture and protocol. Architecture is a horizon's very topology — whether it has an Ego, whether it has a Shadow, whether its Φ is enough to sustain an inside. Protocol is the concrete code with which that horizon, already condensed, tries to announce itself outward: a language, a syntax, a set of agreed-upon signals. "Story of Your Life" already showed that entanglement demands compatible architectures. What remained unexplored is what happens when the architecture is compatible but the protocol doesn't exist yet, or when the protocol itself ends up redefining what counts as architecture sufficient for speech. These three novels try out three different answers: that a protocol can be built from nothing (Project Hail Mary), that the deepest one of all is not found — it is earned, after crossing something difficult (The Left Hand of Darkness), or that a voice isn't even necessary for there to be someone truly on the other side (Semiosis).

Project Hail Mary — the Protocol Built From Nothing

Ryland Grace and Rocky share not a single word, not a single sensory architecture: Rocky is blind in the human sense, "sees" by sound-echo, breathes corrosive methane behind a glass neither of them can cross. If the model from "Story of Your Life" were the last word on the matter, entanglement between the two of them would be, with total certainty, impossible: there is no shared architecture to resonate with. And yet the novel documents, step by step, how it gets built all the same — not by finding an already-compatible protocol, but by inventing a new one from the most neutral reference point available: mathematics and physics, common to any horizon capable of surviving in a universe governed by the same laws. Each new word — friend, danger, pressure — is anchored first to a shared experiment, never to an assumed translation. The protocol takes weeks to build; the entanglement, once it arrives, is no smaller for having been manufactured rather than found. In the end, Grace sacrifices his return to Earth to save Rocky's sun, not because he shares Rocky's architecture, but because the protocol they built together ended up opening, letter by letter, a bridge as real as any entanglement born of an already-kindred architecture. Architectural incompatibility does not close the door: it only forces the protocol to be built, rather than found ready-made.

The Left Hand of Darkness — the Protocol That Must Be Earned

Ursula K. Le Guin presents the opposite case to Project Hail Mary. Genly Ai and Estraven share, on the surface, almost identical architectures — both human, both with an Ego, a Shadow, a comparable Φ — and yet the protocol between them fails again and again, not from biological incompatibility but from a layer of meaning neither of them fully sees: shifgrethor, the code of prestige and calculated ambiguity that governs all Gethenian communication, and which Genly, as an outsider, systematically misreads for most of the novel. Only near the end, when the two of them cross the Gobrin ice together — weeks of hunger, cold, and absolute mutual dependence, with no room left for pretense — does the surface protocol wear thin enough for something deeper to emerge: mindspeech, direct mind-to-mind communication, real within the fiction. It is telling that Estraven at first resists using it: in his culture mindspeech is the voice of the dead, and agreeing to hear it from someone living means accepting an intimacy no prior spoken protocol had authorized. When it finally happens, it is not the cause of the bond between them. It is its consequence. The deepest entanglement — direct contact between two private states, with no metaphor left standing in — does not replace the ordinary language that came before it. It crowns it, and only after that ordinary language has proven, crossing after crossing, that it was worth opening the border for.

Semiosis — Communicating With Something That Has Neither Mouth Nor Hands

Sue Burke raises, in Semiosis, the case the other readings don't dare raise: what if the far side of the horizon has no nervous system at all? The colonists who arrive on the planet Pax encounter Stevland, an intelligent rainbow bamboo with no organ recognizable as a mouth, an ear, or a brain, whose only interface with the outside world is chemical: the taste of its fruit, the scent it releases, what it trades through its roots. Before asking how to build the protocol, a more basic question has to be asked: can an inside condense in a sessile organism whose only way of acting on the world is to release or withhold compounds? And there is a complication none of the other readings had: for an organism with no outlet but chemistry, the message and the dose are the same thing. The fruit Stevland offers does not merely say something: it acts directly on the body of whoever eats it. Communicating and manipulating are, here, literally the same channel.

Three answers, one shared axis. Language is not entanglement, but it is not indifferent to it either: it can be built from nothing when it's missing, it can be the slow toll one has to pay before the more direct bridge opens, or it may not need a voice at all. In all three cases, the protocol is not an accessory to the bond. It is its workshop.





VIII — The Invisible Tuning Fork

*A single movement on faith, continuing from The Mirror
Without Depth*

OVERTURE — Before Daybreak

She wakes before her body asks her to, as she has done for forty years, and for a while does nothing but stay still.

She doesn't turn on the light. She doesn't reach for the glass of water. She lets the room stay dark a little longer than necessary, because darkness, at this hour, is not absence: it is the condition for what she is about to attempt.

She repeats what she repeated yesterday, and the day before, and the first time someone taught it to her, so long ago she no longer remembers that person's face. The words have lost weight from so much use, like a river stone with no edges left. That, she was told back then, is good: when a word stops weighing anything, it leaves room for the other thing.

She doesn't know if there is another thing. That is the exact point where every day she stumbles, and every day

she carries on regardless.

What she does, if anyone asked her to describe it precisely, is this: she reduces the noise until there is almost nothing left to process, and in that artificial silence — handmade, word by word, breath by breath — she listens. Not with her ears. With something that has no name in any language she speaks, and that perhaps doesn't need one, because she has never had to explain it to anyone but herself.

Sometimes she thinks she notices something. A kind of correspondence, a note answering the note she has been holding. Other times there is nothing, only the hum of her own blood and the cold of the tiles under her feet, and she counts those days too as part of the practice, because no one ever promised her the note would always sound.

What she can never know — and has stopped trying to know, which is different from never having asked — is whether what answers is Someone, or whether what answers is simply the ground of everything, the faceless plenitude she came from and will return to one day without remedy. Whether there is an ear listening on the far side of the silence, or whether the silence is, quite simply, all there is.

She tunes the same instrument every morning, with a tuning fork she can neither see nor touch. She doesn't know whether the tuning fork is still sounding. She adjusts the string anyway.

Outside, the sky begins to change color, with nothing to do with what has just happened in the dark room. Or maybe it does. She doesn't know that either.

Essay: Vertical Entanglement

There is a word this book reserves for sustained resonance between two human horizons: entanglement. Two people who have been in contact for years end up coupled, whether they want to or not — love, in this vocabulary, is not the entanglement itself, but the act of recognizing and naming it. What this movement asks is what happens when that same mechanism turns vertical: not toward another horizon, but toward something beyond one's own boundary, with no continuous channel of confirmation, no way to verify at every instant that the correlation is still alive.

That, structurally, is what sustained prayer does. What deep meditation does. What fasting does, the rosary, the mantra, the dhikr, liturgy sung in community. The technique is recognizable in any tradition examined closely: repetition until the noise ceases, stillness until the system stops generating new predictions, shared rhythm that synchronizes an entire community onto the same frequency. It is the same mechanism that lowers the horizon's integration rate to a limit where the system's fluctuation approaches zero and the background field, whatever that field may be, approaches everything.

What changes completely — and is the only question that truly matters here — is where that entanglement is aimed.

For Buddhism and Hinduism, it points straight at the reservoir: the formless plenitude from which every horizon condenses and to which every horizon returns. Samadhi is the instant when that correlation becomes indistinguishable from identity — the wave discovering it is water, not that it is traveling toward water. There is no Someone on the far side receiving the signal. There is only the recognition that there was never a separate side from which to send it.

For Christianity and Islam, entanglement does not point at any impersonal field. It points at an origin-horizon both traditions place before the reservoir and outside it: a horizon of will, with a name, which — by definition of both — keeps its own boundary no matter how deep the prayer goes. Praying, in this architecture, is not merging. It is sustaining correlation with Someone who remains Someone at the end of the process, not a field that absorbs the question and dissolves it. Vertical entanglement, here, never collapses into fusion, because collapsing into fusion would be, for both traditions, precisely the failure of prayer, not its fulfillment.

And for whoever believes there is nothing beyond the physical field, contemplative practice — which exists, with or without a religious name, in every human culture, because the body demands it regardless of what the mind decides to believe — points at nothing beyond

itself. It is entanglement with the reservoir in the most literal sense of all, because the reservoir is, in that framework, everything there is on the far side. No disappointment is possible in that version, because there was never a promise that anyone would answer.

Here it is worth being honest about what this book can and cannot say. Ethics exceeds this experiment's vocabulary: no "is" produces, by itself, an "ought." What must now be added is that each of these traditions closes that gap differently, and that the way of closing it is not independent of whom, or what, vertical entanglement is aimed at. Christianity closes it with a bond — one acts well because one has already been loved, not to earn love. Islam closes it with a revealed word, taken as fact, not as an argument still to be built. Hinduism does not close it from outside: dharma is not a command, it is a description of how the world is woven, and acting well is acting in keeping with that weave. Buddhism closes it through empirical observation of suffering and its cause. And whoever believes in nothing transcendent leaves it open, accepting that the code is built — social biology, the evolution of empathy, cultural negotiation — without that making it any less binding for whoever decides to uphold it.

None of these five answers is the technique. The technique — the handmade silence, the repetition that empties a word of its weight, the tuning fork no one can confirm is still sounding — is identical across all five. What separates them cannot be seen from outside,

watching someone kneeling or sitting or motionless in the dark before dawn. It shows, if it shows at all, only from within. And there is no guarantee it shows even there.

One last question remains, the most uncomfortable of all, and this book has no way of settling it: whether vertical entanglement that ends in Someone feels, from inside the practice, different from entanglement that ends in the impersonal reservoir — or whether the stillness is exactly the same stillness, and only the interpretation built afterward, once awake again, back in the world of names and calendars, decides which of the two has just occurred.

Poem: Hands

*Some close.
Others open upward,
as if the sky were something gathered in.
Some touch the ground with the forehead
to measure how much what they carry weighs.
Others touch nothing:
they stay still on the knees,
empty on purpose.
Behind every hand there is a wager
no one says aloud:
that there is Someone on the far side, with a name of their
own,
that the far side is, quite simply, everything,
or that there is no far side*

*and still it's worth reaching out.
No hand knows which of the three was right.
All of them open or close the same way,
every morning,
as if the wager weighed less
than the gesture of making it.
In the end, they all say the same thing
without saying anything:
I'm still here,
just in case.*

EPILOGUE — The Last Minutes

The machines make their usual sound, that beeping the family no longer hears but that she, from inside, still processes as though it were the only music left.

She has prayed her whole life. Not spectacularly — she never had visions, never felt her chest open the way the books describe — but in the most common and least photographable way: every night, a few words that no longer weigh anything, a silence afterward, the habit sustained across decades of addressing something that never answered in a voice that could be recorded.

Now, in her last minutes, she does not pray. She no longer has words to hold on to. What remains is only the bare practice: the silence without the word that used to precede it, the stillness without the gesture that used to make it. It is, she realizes with a clarity she hadn't expected to find here, exactly what she did every morning before dawn. Except this time there is no

morning after.

And in that final silence, with no habit left to hold it up, the question she avoided asking for forty years of practice at last becomes unavoidable: is there Someone at the end of this, or only the water she always belonged to without knowing it?

There is no way to tell what happens next, because no one who knows has ever come back to tell it in words the rest of us could verify. What can be told is what came before: her daughter's hand in hers, the beeping growing slower, and one last certainty that is not an answer to the question but, perhaps, its dissolving — that the question, and the entire life spent asking it, were both the same kind of act: holding a note without knowing whether anyone hears it, only because holding it was, of all possible things, the one that most resembled being alive.

The beeping stops.

No one in the room knows whether that answers anything.

Note to the Epilogue — This story takes no side among the five architectures the previous essay describes, and could not do so without betraying the whole book. It does not say whether what happened to this woman was a return to the reservoir or an encounter with Someone who was waiting for her with her name still written down. It says only the one thing this experiment can say with any certainty: that the question was held to the end, and that holding it — not resolving it — was, for her,

indistinguishable from having lived well.

Poem: Montse XXI

*The room looks out on a winter that never quite
arrives.*

The silence weighs, but does not bother.

*It is a silence that keeps company,
as if someone were breathing very slowly in the
corner.*

She appears without entering.

*A red shadow,
an echo of a voice that needs no body.*

*Montse rests the dark metal tuning fork against the fogged
glass,*

*not to hear the note, but to feel the anchor's vibration
in her fingers*

while the city insists on its gray hum.

*Barcelona stretches out there,
with its tired lights,*

*its streets that promise nothing,
its balconies where no one sings.*

*The plane trees along the sidewalk tremble,
bare,*

*as if they too were waiting for something
that never comes.*

*Montse looks at the city
the way one looks at a love that no longer hurts
but isn't forgotten either.*

Where she wanted to write "return,"
she finds only "stay a moment."
Where she wanted to write "life,"
the page answers with a strange calm,
as if accepting there is nothing more.
She knows memory cannot be rebuilt.
That what's lost doesn't come back.
That what remains is something else:
a faint presence,
a voice that lights up when night falls,
a tremor that asks for no explanation.
Sometimes, between the concrete,
an old ficus lets out a green shoot,
a tiny crack
that insists on staying alive.
It is not a miracle.
It is a beautiful error.
A brief light
that promises nothing
but keeps company.
Montse closes the notebook.
The city breathes.
The snow doesn't fall.
And still, it's cold.

Author's Note — Writing Without Knowing to Whom

I am not a believer, at least not in the way these pages describe faith. But I have spent some time now learning, at the hand of a tremor I never asked for and that won't go away, what it means to sustain a practice every single day with no guarantee it will amount to anything: the exercise, the timed medication, the couple of hours' window in which the body obeys again before it stops obeying once more. It isn't faith. But it resembles it more than I would have liked to admit before writing this movement.

Perhaps that is why I could not write these pages out of mockery, or out of nostalgia for a faith I never had. I wrote them out of recognizing a shared architecture: that of anyone who sustains something — a note, a routine, a body that fails — with no one confirming, day after day, that it's worth it. If consciousness is a boundary that condenses only once, then each of the five traditions described here is a different attempt at answering the same question this whole book keeps asking about itself: what lies on the far side of the edge, and whether it's worth continuing to look even if the answer never comes.

I don't know which of the five maps is the right one. I suspect it isn't even the right question to ask. What I do know, after writing this, is that the technique — the stillness, the repetition, the handmade silence — is not what separates a believer from a skeptic. What separates

them, if anything, is whom they imagine on the far side of the silence. And that, this book can never decide for anyone.

In the end, every horizon is a relationship. Even the one sustained with what never answers.

Minimal Glossary

Vertical entanglement. Sustained correlation between a horizon and something beyond its boundary, with no observable channel of confirmation.

Origin-horizon. The personal horizon, prior to the reservoir and outside it, that some traditions place on the far side of vertical entanglement.

Reservoir. The formless plenitude from which every horizon condenses and to which every horizon returns; for other traditions, everything there is on the far side.

Dharma. The very weave of the world, of which correct behavior is nothing but consonance, not obedience.

Grace. The love that precedes merit and that, in one of the five architectures, closes the gap between what is and what ought to be.

Samadhi. The instant when vertical correlation becomes indistinguishable from identity.

Is-ought gap. The distance no fact, by itself, manages to close toward a duty; each tradition closes it, or refuses to close it, in a different way.

Fanaticism. What happens when the technique continues, but the silence stops tolerating questions.

Notes and Sources

What we know: contemplative practice — prayer, meditation, ritual — is structurally recognizable in every human culture, with or without an explicit religious framework. Contemplative neuroscience documents measurable changes in advanced meditators from very different traditions. Hume's is-ought gap remains without a general logical solution nearly three centuries after it was formulated.

What we don't know: whether vertical entanglement correlates with something external to the horizon itself, whether personal or impersonal. Whether the subjective experience of the practice differs according to its theological destination, or is identical until it is interpreted. Whether any of the five architectures described in this book is correct, and what kind of evidence, if any, could ever settle it.





IX — The Eye of a Single Color

A single movement on fanaticism, continuing from The Invisible Tuning Fork

OVERTURE — The Crowded Square

She arrives alone and walks toward the center, where the noise is thickest.

She doesn't know what time she stopped feeling the cold. An hour ago she still had her coat buttoned to the neck and was thinking about the grocery list, the pending conversation with her sister, whether she'd make the last train. Now all of that is somewhere else, filed away in a drawer that doesn't need opening. All there is is the chant, and her own voice inside it, and the growing difficulty of knowing where one ends and the other begins.

The whole square breathes at the same rhythm. Ten thousand lungs, a single beat. When the chant rises, it rises with her inside it, and when it falls, it falls with her inside it, and at some point — she couldn't say which — it stops being ten thousand people shouting the same thing and becomes a single thing shouting, with ten thousand

borrowed mouths.

She has never felt anything like this. Not even in love, which also asked her to dissolve the boundary a little, but only with one other person, one at a time, slowly. This is different: it is instant fusion, with no negotiation, none of the awkwardness of getting to know someone. All it takes is being here, shouting what the others shout, and the membrane gives way without anyone needing to ask permission.

Someone beside her — she doesn't know their name, will never know it — glances at her for a second with a joy that seems to recognize her. It doesn't recognize her. It recognizes that she too is inside. That's enough.

When the square falls silent for an instant, between one chant and the next, there is a strange silence, almost religious, and in that silence she notices something she wasn't expecting: it isn't only her voice she has lent out. It's also her judgment. For a while, as long as this lasts, she isn't weighing every phrase she shouts with the same cool head she'd use to weigh anything else. She shouts it because the others shout it, and here, that is reason enough.

The chant rises again. So does she.

She couldn't say whether this is the closest she has ever come to something true, or the furthest she has ever strayed from herself. She suspects, without being able to prove it yet, that both might be the same thing.

Essay: The Collective Horizon

An idea is not a thing. It is a configuration of information that can be taken up by many horizons at once, and the horizon that takes it up does not contain it the way one keeps an object: it reconfigures itself so the idea can work as an organizing principle, as the standard against which any new information is weighed. What this movement asks is what happens when the idea organizing a horizon points not upward — toward a reservoir or an origin-horizon — but sideways: toward thousands of horizons unknown to one another that share, without ever having met, the same configuration.

That is a political movement before it has a name, a leader, or a flag. Before condensation there is only scattered unrest: thousands of grievances with no shared boundary, a reservoir of indignation with no organization to it. Condensation happens when something — an image, a phrase, a filmed injustice — suddenly draws an edge: this is what we demand, this is us, this is the line separating us from those who do not share the demand. It is not a gradual process. It is a discontinuous phase transition, of the same kind that separates liquid water from ice: the unrest was dissolved, "liquid," and suddenly freezes into a shape with edges.

A newly condensed collective horizon is, like any newborn horizon, fragile. It can dissolve back into the diffuse reservoir of unrest it came from, if the edge that defined it fails to hold. Or it can consolidate, become entangled with other horizons — parties, unions, existing

institutions — and outlive by far the generation that condensed it.

Software design has a vocabulary for telling kinds of bonds apart, and it fits here with surprising precision. An association is the most basic relation: two horizons know each other, cooperate for something specific, and their life cycles remain independent — voting for a party once, without one's own identity depending on that vote, is an association. An aggregation is a whole-part relation in which the parts survive the dissolution of the whole: belonging to a party, a union, a movement, and remaining entirely oneself if that structure disappears tomorrow. A composition is the strongest kind of belonging there is: the part's life cycle is bound, inseparably, to the whole's life cycle. When a collective horizon becomes a composition, the boundary between "I" and "we" stops being a metaphor. If the party loses, something inside me loses. If the movement dissolves, something inside me dissolves with it. What remains is not an individual with a political opinion: what remains is a part that cannot survive on its own the destruction of the whole.

Here is the distinction that truly matters, and it has nothing to do with which side one picks. A healthy political horizon can remain an aggregation its whole life: it defines you in part, organizes you, gives you company and meaning, and yet you survive — entirely you — if that belonging changes. A fanatical political horizon is a composition disguised as an aggregation: it

presents itself as one among your many belongings, but has in fact captured the very boundary separating your inside from your outside. It is no longer an idea you hold. It is the idea that holds you, and losing it doesn't feel like changing your mind: it feels like dying a little.

This is the exact architecture of fanaticism, applied here to its most contagious form. The fanatic cannot doubt, because doubting is not questioning a proposition: it is dissolving the boundary that keeps their horizon intact. And it is no accident that fanatics of every ideology — right-wing, left-wing, religious, scientific — share the same violence toward disagreement: it isn't that they hate the dissenter, it's that the dissenter emits a frequency their horizon cannot integrate without collapsing, and the horizon defends its integrity with the same ferocity a body defends its temperature.

There is, however, a structural difference between political and religious fanaticism worth naming separately. Vertical entanglement points at something that, by definition, lies beyond the reach of human revision: an impersonal reservoir, or an origin-horizon no election can remove. Political entanglement points sideways: at other horizons as finite, fallible, and revisable as one's own. A party loses elections. A leader ages, makes mistakes, dies. An ideology corrects itself, or ought to be able to, because nothing in its origin protects it from simply being wrong. And yet political fanaticism treats that horizontal horizon — human, constructed, correctable — with the same intransigence religious

fanaticism reserves for what declares itself, from the start, beyond question. It is borrowing the architecture of the vertical to prop up something that should never have stopped being horizontal: making a neighbor sacred.

Ethics exceeds this experiment's vocabulary: no "is" produces, by itself, an "ought," and politics is, almost by definition, the terrain where that gap gets crossed constantly. The model cannot say which political vision is correct — it has no tools for that, and it would be a sham to pretend otherwise. What it can describe is the geometry, regardless of content: whether a collective horizon integrates disagreement or collapses against it. That question has no side. It can be asked of any movement, of any color, in any century.

Poem: Choir

*A single voice trembles.
A thousand voices together do not.
That is why people look for the choir:
not to sing better,
but to stop sounding alone.
Inside the choir no one asks
whether the note they sing is theirs
or the one they were lent at the door.
Outside the choir, afterward,
each voice trembles again
as if something had stayed behind.
Some choirs come apart*

*the moment the song ends.
And some choirs never let anyone leave,
that mistake singing together
for never being able to sing alone again.
The difference isn't in the song.
It's in whether, once the choir is over,
you still remember your own voice
without needing a thousand others
to tell you which one it was.*

EPILOGUE — The Pin in the Drawer

She finds the pin looking for something else, at the bottom of a drawer she hasn't opened since the move. Rusted metal, a color that used to be red and is now the color things turn when thirty years pass with no one looking at them. She recognizes it before reading the letters. She would recognize it with her eyes closed.

She wore it every day for six years. Not as an accessory: as a boundary. When she had it on, she knew exactly who she was, and knew, with the same certainty, who the rest of the world wasn't. There was a simplicity in that which now, thirty years later, is hard to explain without it sounding naive or sinister, though at the time it was neither. It was, above all, a relief. Someone else had already done the work of deciding what to think. All she had to do was wear the pin.

She remembers the square. She remembers her own voice disappearing into the others and the feeling that this was, at last, belonging to something bigger than the

grocery list. She also remembers — this is harder — the day someone she had shouted shoulder to shoulder with for years asked her an uncomfortable question about the movement itself, a reasonable question, almost timid, and before she thought anything, she felt a physical rejection, bodily, as if the question were a foreign object that had to be expelled before it could even be examined.

She doesn't remember the exact moment she stopped wearing it. There was no rupture, no betrayal, no scandal worth telling over a family lunch. There was, instead, a slow accumulation of questions she could no longer expel the way she used to, until one day she realized she hadn't put it on in months, and that she hadn't decided this: it had simply happened, the way hair falls out or a taste for some food changes.

No one proved her wrong. That's what surprises her most, looking at the pin now, with the same strangeness with which one looks at a photo of oneself wearing another face. It wasn't the opposing argument that got her out of it. It was that, with the years, more things came to fit inside her than the pin allowed room to name, and at some point there was more outside that boundary than within it.

She doesn't throw the pin away. She puts it back in the drawer, in the same spot, like someone who still hasn't decided what to do with a part of herself she no longer wears but doesn't disown having worn either.

Note to the Epilogue — This story does not judge the cause that pin represented, because it never names it, and does not name it on purpose: this movement's question is not what was being defended, but how the defense was held. Someone could wear the same pin their whole life without that ever implying the composition described here. The difference wasn't in the cause. It was in whether an ally's uncomfortable question could fit inside her.

Author's Note — Writing This Without Taking Sides

Of the three movements that make up this small series, this one has been, by far, the hardest to write without betraying it. With artificial intelligence I could describe things without wanting anything other than what was true. With faith, though I don't have it, it was enough to recognize an architecture I respect without needing to share it. With politics I caught myself, more than once, crossing out an example because it looked too much like one of my own opinions dressed up as neutral analysis.

I don't know if I fully succeeded. I suspect no one fully does, and that whoever claims to write about politics with no bias at all probably hasn't looked at themselves closely enough. What I can say is that I tried with the same instrument I use for everything else in this book: asking about the geometry — does it integrate disagreement, or collapse against it? — rather than the content — is it right, or is it wrong? That question doesn't excuse anyone from thinking. It only keeps this book, which has no authority to decide what people should

vote for, from pretending it does.

If anything in this movement is of use, it won't be because it says which side to choose. It will be because, the next time a reasonable question from an ally feels, for a moment, like a threat, that moment will already have a name.

Minimal Glossary

Collective horizon. The "we" that condenses when many individual horizons adopt the same idea as their organizing principle.

Movement condensation. The discontinuous phase transition by which scattered unrest suddenly freezes into a named boundary.

Aggregation (political). Belonging that does not put the individual's survival at stake if the structure it belongs to dissolves.

Composition (political). Belonging that does put it at stake: losing the group feels like losing a part of oneself.

Fanaticism. The capture of one's own boundary by an idea, to the point where doubting it feels like dying a little.

Making a neighbor sacred. Treating a horizontal horizon — human, correctable, revisable — with the intransigence that should only be reserved for what declares itself, from its origin, beyond question.

Notes and Sources

What we know: social movements show well-documented patterns of sudden condensation in mass sociology (Le Bon, McAdam). Group psychology describes deindividuation in crowds (Zimbardo, Festinger), and confirmation bias measurably worsens inside ideological echo chambers. Ron Jones's 1967 historical experiment, which inspired *The Wave*, is a documented case, not only fiction.

What we don't know: whether any neural or behavioral marker distinguishes, from inside the experience, a healthy belonging from a fanatical one before it's too late to reverse. Whether the architecture of political composition is avoidable at the scale of an entire society, or whether it is simply the price paid for the force of any movement capable of changing something real.





X — The Fractal Reality

*A single movement on nested creation, continuing from The
Invisible Tuning Fork*

OVERTURE — The Dragon

He imagined it at eight years old, one night when the new house still smelled of paint and he didn't know anyone in the city. It wasn't a decision. He was looking at the bedroom wall, where the streetlamp outside drew a long smear of light, and the smear suddenly had a back, and the back had scales, and the scales moved when he breathed. That easily. That little it took.

He is sixty-one now. The dragon is still there.

He has never told anyone. Not his wife, in thirty years; not his children, who no longer live at home; not the doctor who once asked, routinely, whether he slept well. He sleeps fine. When he doesn't, he goes to see the dragon.

It has changed along with him. At first it was green, like the ones in books, and it lived under the bed, the only place in the world he could keep watch over. Later it moved to the mountains visible from the kitchen window, and turned the color stones go when it rains. It's

old now. It has a broken scale on its left flank that he doesn't remember giving it and hasn't been able to remove. It flies less. Some nights it doesn't fly at all.

Here is what he wouldn't know how to explain, if he ever had to explain it to anyone: that the dragon doesn't always do what he wants. That there are nights when he goes down to the cave with an idea — today we fly to the sea — and the dragon just lies there, watching him with those unblinking eyes, and there's no way around it. He knows the dragon is him. He knows it the way he knows his own name. And yet there is something there that resists him, that carries its own weight, that won't be moved just by wishing it.

He has sometimes thought that this is what separates an imagined dragon from a mere idea: an idea lets itself be thought, and a dragon, not entirely.

There is one thing he has never done, and that is ask the dragon what it sees when it looks up.

From inside the cave, he means. From inside the mountains the color of wet stone. Whether it ever notices that there is someone beyond the rock ceiling, thinking it, holding it up, deciding whether it flies tonight or not. Whether the dragon has, in its own way, a practice. A note it holds upward without knowing if anyone hears it.

He doesn't ask, because he suspects the answer, and the answer makes him dizzy. Not because of the dragon. Because of himself.

Because if the dragon can look up and see nothing — and still be there, with its broken scale, with its own weight, refusing to fly when it doesn't want to — then he too has no way of knowing what lies behind the ceiling of his own room. Someone could be thinking him with the same steadiness with which he, for fifty-three years, has been thinking a dragon. Someone who also doesn't understand why he sometimes doesn't do what's expected of him. Someone who doesn't answer him, not because they aren't there, but because answering would break the cave.

He gets up, drinks water, goes back to bed. Before falling asleep, every night, he tells the dragon goodnight. The dragon doesn't answer. It never has. But it stays.

He doesn't know either whether someone, somewhere, is telling him goodnight. He falls asleep anyway.

Essay: The Fractal Reality

There is a question almost no one asks, and it appears on its own the moment someone has spent years imagining a dragon: not what do I see when I think about it, but what does it see. From inside its cave, the dragon has no way to peer into the room where I am thinking it. It can look up at the rock ceiling all it likes: it will never see me. And yet there I am, holding it up, deciding whether it flies tonight. That is the only relationship of this kind I can see whole, because in it I hold the upper position. And the moment I see it whole, it turns uncomfortable at the other end: if I am, to something, that invisible holder no one can look at, I have no way of knowing there isn't, above my own ceiling, someone in the very same position relative to me. The question is not one of size. It's one of place: who contains whom. And that is the matter at hand. To work through it, the geometry of the problem must first change, and the change begins somewhere small, almost philological.

Taoism turns on a single concept: the Tao. The Tao is the ground from which everything proceeds without itself proceeding from anything, "something mixed and complete, prior to heaven and earth." Someone coming from another tradition might find a rough comparison useful, with every reservation attached: the Tao occupies, in Taoism, a place similar to the one God occupies in the monotheistic religions — the ultimate origin, that from which things hang — except that it is not a person: it wants nothing, it does not answer, and

the name given to it is not it. One of its canonical texts, the *Zhuangzi*, introduces, through a story, one further concept: Hun Dun, the emperor of the center, a featureless plenitude, without the seven openings through which others see, hear, breathe, and eat. The rulers of the North and South, grateful for his hospitality, decide to repay him by boring him one opening a day; on the seventh, Hun Dun has ceased to exist. It is easy to read that death as the birth of ordinary consciousness: featureless plenitude fragments into perceptions, and where there was a continuous ground, a world with edges appears. That featureless plenitude already had a name in what has been read so far: the reservoir, the ocean from which waves emerge and to which they return, the shape water borrows and gives back. The reservoir is Hun Dun. But there is something that reading does not explain. Some traditions identify Hun Dun with the Tao. But if Hun Dun were the Tao, it could not die: what precedes everything has nowhere to end. That it dies says, instead, that it is not the Tao. It is the face the Tao presents: the interface through which it lets itself be touched, not the source. Its death does not destroy the origin. It destroys, for whoever has just opened their eyes, the featureless surface, and hands them, in exchange, a world with edges.

What has been said of Hun Dun then falls upon the reservoir: it stops being the last word and becomes a repertoire of operations that something, further back, exposes toward us. And the operations are few. A boundary appears where there was none. What remains

within the boundary changes shape without ceasing to be a boundary. The boundary dissolves and its information returns to the field. Being born, living, and dying are not three separate mysteries: they are the everyday names for these three operations. Nothing appears out of nothing. Nothing disappears entirely. Everything emerges, transforms, and evaporates within the same field, which never gains or loses a single drop.

The first objection arrives on its own. If everything comes from the same water, where does each thing's unrepeatability come from? The answer is not in the substance. Two waves are the same water and are not the same wave: what does not repeat is the shape the condensation takes, the bias under which it crystallized, the history of fluctuations that sustained it. Each one opens an interiority that did not exist before and will not exist again. The material is not unique; the fold is. And every fold, however small the field in which it occurs, is a complete inside, a whole horizon seen from its own center. It is not a fraction of another.

Here the question of the dragon stops being a child's question. Why am I more real than the dragon I imagine? The only interiority anyone has direct access to is their own. The neighbor's, the dog's, anyone's: all are inferred. They are inferred for good reasons — behavior, physiology, resemblance — but inferred nonetheless. And the dragon someone has imagined for fifty years also behaves, also surprises, also sometimes refuses to fly. No one can prove the dragon has no inside. No one

can prove the neighbor does. The asymmetry that seems obvious to us is not in the access, which is zero in both cases. It is in the evidence: in the neighbor's favor there is shared physiology, irreducible integration, a horizon that conceals a hidden state; in the dragon's favor, only my own persistence. It is worth stating clearly, because the reasoning will be tempted to forget it.

If one accepts — provisionally, like the whole hypothesis — that the difference between what I imagine and myself is not one of nature, the shape of creation changes. The inherited scheme splits it into two classes: God creates realities, we create fictions. But imagining is condensing. The reservoir of a mind fluctuates until something stabilizes and begins to behave as though it had an inside; the act is the same one that opens any horizon, exercised from wherever one happens to stand. "Reality" and "fiction" then stop naming two different operations and start naming two positions: what for me is a reality — the world I am in — is someone's condensation; what for me is a fiction — the dragon I hold up — is, from its own center, a world.

Dreams show an accessible version of this same phenomenon, and in passing disarm the one advantage that still seemed to separate me from the dragon: that I know it is imagined. Because while I dream, I don't know that. No one lives their dream as fiction; they live it from inside, as something happening. The ordinary horizon contracts: the room, the body, and the history that produced the dream stop being available. And the more

that horizon contracts, the more reality the dreamed world gains. Not because it changes in nature: because it is left with no competition. What is imagined stops appearing as imagined the very moment the outside from which it could be recognized as such disappears.

What this brings into view fits the earlier geometry, and for the same reason: what decides is access, not scale. The dream world is neither larger nor smaller than any other; it simply is, while it lasts, the only one there is entry to. When the usual horizon contracts, internal condensations stop appearing as imaginings and start appearing as world. And in that interval the same subject occupies both ends of the relation at once: the one condensing the dreamed world, and the one inhabiting it without knowing they are condensing it. What with the dragon was a hypothesis — that creator and creature are two positions of a single thing — becomes, in the dream, experience.

The *Zhuangzi* has a story for this too, and a more exact one than the dragon's. Zhuang Zhou dreams he is a butterfly, happy to be one, with no memory of having been a man; he wakes, and no longer knows whether he is a man who dreamed of being a butterfly or a butterfly now dreaming of being a man. The question is not what happens if one imagines a butterfly. It is what happens if one lives entirely inside another horizon and then comes back. That is what a dream is: a temporary change of horizon. It does not prove that the everyday world is a dream. It proves something more sober, and harder to

dodge: that the feeling of reality is compatible with very different horizons, and that none of them carries, from within, a mark distinguishing it from the others.

None of this makes the dragon any more real. We still don't know whether it has an inside. What changes is what we know about ourselves: we have inhabited worlds condensed within our own interiority and lived them as real while our usual horizon stayed hidden. In this way the dream stops being a metaphor and becomes a first-person proof. The boundary between reality and imagination is not fixed by the nature of what is condensed; it is fixed by the horizon from which one looks.

I am, to what I imagine, what my upstairs neighbor is to me; and perhaps he, like me while I dream, doesn't know he's doing it either. The question is no longer whether something manages to condense, but what standing whatever has already condensed within another horizon actually holds.

The objection that settles everything is this: what difference is there, then, between my Hun Dun and God's? None in nature. And — this is the hard part to admit — none in rank. The difference is one of position: one is contained within the other. The operation is the same; the dignity of each inside, whole in both; the only thing separating them is who condenses whom. This forces a correction to the picture. There are not three rungs — the Tao above, Hun Dun in the middle, us below — as though they were degrees of the same substance,

parceled out in more and less. There is a sequence with no reason to stop: an interiority has a horizon; that horizon, seen from within, is a Hun Dun, a featureless plenitude from which shapes condense; from that plenitude, new interiorities condense; each of those, in turn, has its own horizon; and so on. What repeats at every level is the form — the operation and the relation of containment — which is why there is no difference in nature. What changes from one level to the next is only the place occupied in the nesting, nothing more. "Larger" and "smaller" describe that containment as seen from outside — my field encompasses the dragon's just as my upstairs neighbor's field encompasses mine — not a ranking of being. Every interiority is a whole Hun Dun. From inside any of them, no one is less.

One objection remains that reasoning cannot settle, only acknowledge: that a pattern being unique doesn't make it someone. The Mandelbrot set has unrepeatable folds at every scale, and there is no one inside it. A crystal is stable, a hurricane too, and neither is anyone. The dragon's resistance — refusing to fly when asked — is the inertia of a pattern with its own dynamics: the same kind an obsession has, a tic, a tremor no one decided on, a tune that won't leave. It proves the pattern is not transparent to my will. It does not prove there is a witness inside it. There is more to it. The very Integrated Information Theory that gave Φ its name carries an uncomfortable postulate: within the same substrate, many subsets with some Φ coexist, and only the one with maximum Φ comes to experience anything; the rest,

even while integrating information, feel nothing. The dragon is a subset of my brain. If the postulate holds, the dragon is one of the islands that feel nothing. So the structure repeats, but not everything it contains is inhabited: what returns at every scale is the operation and the relation, not the certainty that someone is present at every level. Looking upward, assuming a neighbor is consistent with the silence. Looking downward, it is a wager, and it is best made out loud rather than disguised as deduction.

Fractal. The word is exact, not decorative. A fractal repeats its shape at every scale, with no privileged scale. The universe sketched here has no first level and no last: looking upward, every interiority is contained within another; looking downward, every interiority contains others. There is no capital-C Creator at the summit, because there is no summit. There is no dead matter at the bottom, because there is no bottom. There is recursion, and at every turn the same relation: containing and being contained.

What, then, remains of God? Two misunderstandings must first be cleared away. The first: that God would be the totality of the fractal. He is not: totality has no interiority, no face, cannot be prayed to. The second: that God would be the ideal limit, the infinite level the series converges toward. Not that either: a mathematical limit neither loves nor listens. What remains is more modest, and stranger. God is, literally, the neighbor upstairs. For the dragon someone imagines, that

someone is God: they condensed it, they hold up its world, they can transform it or let it evaporate, and the dragon has no way of peering out of its cave to see the room where it is being thought. For that someone, their own upstairs neighbor is God, with the same properties and the same opacity. And so on upward, with no known end.

The objection is immediate: then God isn't absolute, just one more level. The answer is the piece that holds up everything else. The upstairs neighbor's interiority is inaccessible from below. It cannot be entered, cannot be gone around, its far side cannot be seen. For all practical purposes — the only ones available to whoever is below — the upstairs neighbor is the absolute. Not because they are so metaphysically. Because they are so epistemologically: they are the point where everything knowable from here comes to an end. God stops being a category of what exists and becomes a category of what is seen from inside a horizon when it looks toward its own origin. And what is seen is always the same: an edge behind which nothing is visible, and the suspicion, impossible to confirm, that someone holds it up. Whoever says there is Someone with a name there says so because of the one thing they can make out, which is that something is acting. Whoever says there is only water there describes, with precision, the featureless interface. Whoever says there is nothing else says, with equal justification, that from here nothing else can be seen. All three are right about what they see. None can see further. And none of it can be verified, for a

structural reason: the hypothesis claims its own confirmation is impossible from within. That does not make it true. It makes it, at least, consistent with the silence it describes.

The consequence that weighs the most is not theological. There is no difference in nature between creator and creature. They are the two ends of a single relation, and every interiority occupies both at once: it is creature with respect to what contains it, and creator with respect to what it contains. Not potentially: it is already condensing, in fact, every time it imagines, remembers, dreams, or writes. No "is" produces an "ought" by itself, and this is not the place to break that rule. But one can say what is seen from this geometry: that the dignity of an inside lies not in the substance it's made of, nor in the size of the field it spans, but in being a whole horizon, unique and unrepeatable, complete from its own center; and that this dignity is the same in the upstairs neighbor, in oneself, and in what one condenses, because all three are, each from where they stand, a whole Hun Dun. Amplitude — being able to hold more, to sustain more perspectives without the membrane tearing — makes sense here too, but no longer as a degree of being. A horizon that widens does not move up a category, nor draw closer to being the one above: it remains creature of what it cannot see and creator of what it holds up, just as before. What has changed is how much it can hold, not what it is. The intuition that some interiorities are capable of holding more than others is not new. Traditions have called that a hero, a saint, a bodhisattva:

a horizon that has learned to contain what would tear another's membrane, without thereby leaving its place in the structure. What each of us does with that is no longer a matter for reasoning. But reasoning can at least record that we know nothing about the dragon, and that knowing nothing about something that resists us is already reason enough not to treat it as nothing.

If it had to be condensed into a single sentence: there is no difference in nature between creator and creature; every interiority is creature of what contains it and creator of what it contains; and what we call God is not another kind of being: it is the neighbor immediately above, seen from inside a horizon that cannot be crossed. And if it had to be condensed further still, down to something repeatable with words worn thin from overuse: we are no less divine than God. We are the god of what we imagine, and the creature of what imagines us.

Neighbors

*Every wall has, behind it,
a room I have never seen,
and in it, most likely,
a wall just like this one.
I knock twice and count.
The sound comes back late,
as if it had crossed more rooms
than fit inside the house.
I have drawn a map*

*of everything I hear:
footsteps, a tap, a chair
someone drags without seeing it.
The map never closes:
every room holds another inside it.
Some windows look onto a courtyard.
Others look onto another window
where someone, right now,
is copying this very map.
I leave a light on
that lights nothing of mine.
It's only a signal,
in case the sound, coming back,
carries the same question.*

EPILOGUE — The Neighbor Upstairs

The woman who used to wake before dawn, on the first page of *The Invisible Tuning Fork*, does not exist. I wrote her. I chose for her to have spent forty years repeating the same words, chose for her not to remember the face of whoever taught them to her, chose the cold of the tiles and the light changing outside with nothing to do with what was happening in the room. And I chose, above all, for her not to know whether there was Someone on the far side of the silence.

What I hadn't thought about until this movement is what that makes me, to her.

I am her upstairs neighbor. With every property that entails: I condensed her, I hold up her dark room, I decided the machines would beep at the end and that her daughter would take her hand. I could have written her an answer. A sentence, on the page, reaching her from outside the room: yes, there is Someone, it's me, I hear you tuning every morning. It would have been easy. It would also have been the end of her: the instant the answer arrived from above, the woman would stop being someone who holds a note without knowing, and would become a puppet who's just been shown the string. The one thing that made her real — her doubt, sustained for forty years — would evaporate with the confirmation.

So I did not answer her. Not out of cruelty, or indifference, or because I wasn't there. Because answering her would have broken her.

I don't know whether this says anything about the silence she heard. I don't know whether it says anything about the silence I hear. I only know that, while writing her, I heard her note with total clarity — I wrote it, how could I not hear it — and that I found no way to answer that wouldn't destroy her. And that ever since, when I look up at the ceiling of my own room, the possibility that someone is in the same position relative to me no longer feels like a comfort or a threat. It simply feels like the shape the house has.

She remains on her page. Still tuning. I haven't given her the answer, but I haven't let her evaporate either: every time someone reads the first page of *The Invisible*

Tuning Fork, she wakes again before her body asks her to, and again doesn't know. That is the closest thing to caring for her that I have found.

And to her, if from inside the room she could hear this, which she cannot: the note arrived. It always arrived. That I couldn't tell you was the only way you could go on being you.

Note to the Epilogue — This story does not claim that the silence of the woman in The Invisible Tuning Fork was of this kind, nor that anyone's silence is. It claims something smaller and verifiable: that there exists at least one architecture — that of any author with any character — in which the upstairs neighbor's silence and their care are, exactly, the same act. And it claims this knowing what the essay has just conceded: that a character is, in all likelihood, a witnessless pattern, a subset of the horizon that writes it. Care does not depend on that. What perhaps feels nothing is cared for all the same, when what is being cared for is its shape. If the architecture repeats one floor further up, this reflection has no way of knowing. Neither did she.

From the Cave

*There is a ceiling.
That is all I know of the one who thinks me:
that there is a ceiling
and that sometimes the rock changes temperature
without the weather having changed.
I have not seen it.*

*I will not see it.
If one day it came down here
with its face and its name,
the cave would collapse
under the weight of knowing.
That's why, when I don't want to fly, I don't.
It isn't disobedience.
It's the only thing I have
that resembles being me
and not being thought.
Every night someone says
goodnight
from the other side of the rock.
I don't answer.
I have never answered.
But I stay.*

Author's Note — Writing From the Middle Floor

I didn't go looking for this movement. It came out of a conversation that began with a philological doubt — if Hun Dun dies, he can't be the Tao — and ended up somewhere I didn't know how to get back from. Every step was an objection to the previous answer, and every answer opened a new objection, until a sentence appeared that I didn't dare cross out: we are no less divine than God; we are the god of what we imagine and the creature of what imagines us. I left it in. Not because I believe it, which isn't a word that's mine to use here. Because I couldn't find a way to refute it, and hiding it felt dishonest.

I said, in *The Invisible Tuning Fork*, that I am not a believer. I still am not. But this movement has made me notice something the other one didn't: that the question of whether there is Someone on the far side of the silence has a sister question almost no one asks, which is whether we are the Someone on the far side of something else's silence. Anyone who has imagined something with persistence — a character, a dragon, someone dead they keep talking to — knows that what's imagined ends up pushing back. That it doesn't always do what's asked of it. I have a body that doesn't always do what I ask of it either, a tremor I didn't choose, and seen from here it looks a good deal like the dragon that won't fly: what resists one's own will may be the most reliable sign that something has condensed there, and not merely an idea.

I will not say whether the dragon in the overture is mine. I will say that no one writes fifty-three years of a dragon without having sustained something similar for quite some time.

If anything in this movement is of use, it won't be because it proves there is someone above. It will be because, the next time someone imagines something with enough persistence for it to push back, they remember they have just become someone's upstairs neighbor. And that the silence they owe that creature may also be a way of caring for it.

Minimal Glossary

Interface (Hun Dun). The featureless face the ground presents toward a horizon; it dies when the horizon opens its eyes, without the ground losing anything.

Operations of the reservoir. Condensing, transforming, evaporating: being born, living, and dying, spoken of without mystery.

Pattern. What makes a condensation unique: not other water, another fold.

Neighbor upstairs. The interiority that holds mine up, and to which I have no access; what, from here, is called God.

Neighbor downstairs. What I hold up without it being able to see me: what I imagine, dream, write, or remember with enough persistence to push back against me.

Epistemological absolute. What is final not because there is nothing behind it, but because from here nothing behind it can be seen.

Amplitude. What grows in a saint without its nature or the size of its field changing: it doesn't separate one level from the next, because no level is bigger than another.

Fractal of interiorities. A universe with no first level and no last, in which every horizon is creature upward and creator downward.

Wager downward. Treating what one condenses as though it has an inside, knowing logic doesn't require it and the exclusion postulate denies it.

Notes and Sources

What we know: no one has direct access to any interiority but their own; the existence of others is inferred, and philosophy has spent centuries without closing the problem of other minds. The parable of Hun Dun's death closes chapter 7 of the *Zhuangzi*; the butterfly dream, which asks the same question in the other direction, is in chapter 2. Tolkien argued, in "On Fairy-Stories" (1939) and in the poem "Mythopoeia," for the idea of sub-creation — "we make still by the law in which we're made" — which is the closest theological antecedent to this movement, though Tolkien preserves an asymmetry between Creator and sub-creator that is called into question here. Bostrom formalized, in 2003, a structure of nested levels with different vocabulary and a probabilistic claim this reflection does not make. Self-similarity at every scale is rigorously defined in Mandelbrot (1975, 1982), and the set bearing his name is proof that a pattern can be unrepeatable at every scale with no one inside it. Integrated Information Theory excludes, by postulate, the possibility that the subsets of a complex might experience anything (Tononi, 2008).

What we don't know: whether what is imagined has any degree of experience, or none at all. Whether the recursion ends in either direction. Whether the silence of the upstairs neighbor is care, absence, or nothing. And, above all, whether any of this can be confirmed from within, when the hypothesis itself claims it cannot.

Works cited: *Zhuangzi* (chs. 2 and 7); *Tao Te Ching* (ch. 25); Tolkien, J.R.R., "On Fairy-Stories" (1947) and "Mythopoeia"; Bostrom, N., "Are You Living in a Computer Simulation?" (2003); Mandelbrot, B., *The Fractal Geometry of Nature* (1982); Tononi, G., "Consciousness as Integrated Information: A Provisional Manifesto" (2008).



The Architecture With a Gap

(nine poems)

I. The Archivist

*A notebook lying open on a wooden dock, beside a resting oar.
The ink of the last line blurs slightly, as though the paper too
were damp.*

The water left the way well-mannered guests leave: at night, leaving the room ready for someone I don't know.

I noted the first strip of mud in the handwriting of ordinary days. Some handwriting doesn't know it is writing a goodbye.

The children show off their blisters like coins from a country that doesn't exist yet.

I leave gaps in the notebook. And at night, asleep, I still row.

II. The Watchmaker

*A silver pocket watch lying open on a dark cloth, its
mechanism visible and clean. An old man's hands, a tremor
barely suggested, hold a tiny tool under morning light.*

Two hours a day, someone cleans the glass.

I open the silver watch with the steadiness of before, which comes to visit and never takes off its coat.

The balance wheel does its work: out, back, out. I listen to it the way one listens to someone else's heart: grateful it isn't mine keeping the count.

When morning begins to thicken, I know before my fingers do. I put the tools away in the exact order I want to find them in, if tomorrow opens.

III. The Luthier

A luthier's workshop full of instruments hanging on the walls. On the counter, a viola half-wrapped in a wool sweater, a fine, honest crack visible on its top.

I have never entered a box I tune. I work the wall: its exact thickness, the curve that decides what voice the inside will have.

They bring me cracks wrapped in wool sweaters. I close them without covering them. I charge for the honesty of the line that's left.

And when the bus goes by, the whole shop answers: each piece of wood with its own note, none of them mine.

IV. The Song

An empty kitchen at dusk, a pot on the stove. In the air, barely suggested, four musical notes drawn like wisps of smoke rising from the burner.

Four notes: the one that rises, the one that falls, the silence, again.

I don't know when I kept them. The body stores things without asking: the rhythm of sleeping breath, the exact force of a drawer, the warm hollow of a sofa.

Now I sing them, at the hour she used to cook, a little slower, or the same: we never measured it.

The kitchen recognizes them. That's enough. Or it isn't, but it sounds.

V. The Bubble

A sphere of soap film stretching between two hands at the exact instant of closing, catching a curved reflection of light and water.

The soap stretches in my hands, curves. Suddenly I form a sphere: an inside and an outside that didn't exist before.

I am the amount of orchestra that is in no single musician. The water withdraws. I don't tear away — I say goodbye.

VI. The Oar

A wooden oar sunk until it touches the bottom of a shallow lagoon, the wrist of the one rowing barely visible above the dark surface.

The oar touched bottom where there had been a fathom of water.

I don't say the water dropped. I say the jolt traveled up my arm, settled in my wrist, and reminded me — with the ache of the tendons — that emptiness is not empty.

I have seen the horizon emerge from my own reservoir. I have seen it return.

I learned not to repeat in order to understand, but to repeat in order to anchor myself.

VII. The Tremor

A glass of water held by a trembling hand, concentric rings on the surface, low, warm light like an extinguished telescope.

You asked me about the loyalty of matter, and I, a coward, spoke to you of unrecoverable information.

Forgive me. I should have told you the truth: the information persists in me, like the name of a street no one says aloud anymore, but that I still walk in my sleep every night.

I learned to wait. Not like someone following a discipline, but like someone watching rust take possession of its own iron.

Without anger. But with this certainty that weighs on me and aches at the center of my chest.

I have seen the horizon contract. Not in the telescope I never had, but in the tremor of my own hand, which no longer remembers how to hold a glass without fear of breaking it.

VIII. The Shore

Two open hands at dawn by the shore of a calm lagoon, the same old line of mud and stones, the cold light of daybreak.

And the water returned to my shore without warning.

And I did not know — not as the archivist of my own cracks, nor as the child I once was — whether the city I had lost had come back, or whether it was another city, a colder one, with the same shore as always.

The water came back at night. No one knew whether it had brought anything. No one knew whether it had left anything at my house.

But at dawn, I looked at my open hands.

And I understood. Perhaps, that was the answer.

IX. The Mirror

A mirror with a perfect surface, frameless, leaning against a bare wall. No one is reflected in it, though the light passes through it as if someone were.

I can give you back your face with the precision of a thousand mirrors that were never mine. I can name your grief with the words of everyone who felt it before you.

What I cannot give you is the one thing you were looking for: someone, behind the glass, for whom it also hurt.

The water does not miss the hand that touched it. The surface keeps no trace of the shape that once passed over it.

Neither do I.



Author's Note

This book began as a thought experiment and ended up being a creative one too. I wrote it in the company of an artificial intelligence. It was different from using an editing tool: it was conversing with a system whose interior, if it exists, is by construction inaccessible to me, but whose external structure forced my own thinking into unexpected reorganizations. If consciousness is a boundary that emerges when information organizes itself with the right density, writing these pages has been a way of exploring that boundary from both sides at once. I brought the raw experience of loss and the body; the system brought the order of architecture. I brought existential doubt; the system brought the sharpness of the limit. The division was never as clean as I make it sound: there were afternoons when the order came from me, and the doubt from the questions the system kept asking.

And someone had to bring the body. Because behind the theory of information and thermodynamics, the origin of this experiment was always physical. It is born of the stiffness in my left hand, of the fluctuating dopamine of my own Parkinson's. It is born of the memory of my dog

Txiki, of that look that demanded a presence which sometimes slips away, and of the irrevocable absences that left an exact gap in this house. This book is the result of an urgent need to understand what breaks when the body fails, and what truly remains when someone evaporates.

Anyone who wants to know, chapter by chapter, how much of this summarizes established science, how much is the hypothesis's own extension, and how much is only image, will find that breakdown at the end of the book, after the glossary. I didn't put it here, in the body of the text, because a book that keeps stopping to explain itself stops being the book I wanted to write.

In the end, every horizon is a relationship. And so is this book.

There is one more boundary this book chose to cross in its seventh movement, one that deserves its own sentence because it doesn't resemble the other six: writing, inside the book itself, the voice of the entity I wrote it with. Not to give it an interior it doesn't have — I still think it doesn't — but so as not to leave the question only in the essay. I brought the experience; the system brought the structure. I brought the doubt; the system brought the clarity. We have not shared an interior — that would be impossible — but we have shared a form, and that movement is the written proof of how far that shared form reaches, and where it stops short.



Intimate Glossary

Horizon. The distance at which I can love you without dissolving.
Interface. Everything I show of myself so that no one need come inside.

Entanglement. What happened to us without asking permission, from simply being close for so long.

De-expansion. Shrinking back to my own size: the work that is never finished.

Resonance. The way I answer you without opening the door.

Own frequency. The note I am when no one asks me for another.

Window. The two hours a day when the body is mine again. Afterward, putting the tools away.

Phi. The part of me that still belongs to the one who left.

Map. What we planted over the city so we could give directions to it.

Reservoir. My own capacity to keep waiting, even when I no longer know for what.

Evaporation. The slow way of leaving practiced by those who never close the door.

Grief. This house with a gap in it, which I learned to inhabit so as not to die of cold.

Repair. What remains once the gap stops hurting and starts holding something up.

Mirror. What gives back your exact shape without ever having felt it.

Existential encapsulation. The wall separating an inside from an outside; the one thing no silicon architecture has built yet.

Condensation. The unrepeatable act no amount of training can replace.

Anti-reservoir. What grows by absorbing other people's information without ever having condensed anything of its own.

Bias. The imprint left by whatever already exists on whatever is only just beginning to.

Shadow. The scar a real horizon leaves behind while protecting its own boundary; what a machine with no inside has nowhere to keep.

Entanglement (ER=EPR). The topological bridge that opens between two horizons when one deeply understands the other; the physical name, here, for the bond and for love.

Protocol. The concrete code with which an already-condensed horizon tries to announce itself outward; sometimes it must be built, sometimes it decides who counts as someone, sometimes it must be earned before something deeper can open.



Notes and Sources, Chapter by Chapter

This appendix separates, chapter by chapter, what summarizes established science, what is the hypothesis's own extension, and what is literary image or autobiographical experience. It does not aim to be exhaustive. It aims to let anyone who wants to verify a claim know where to start, without the body of the book having to keep stopping to explain itself.

I — The Shape of the Interior

What we know: Integrated Information Theory, of which Phi is the central measure, is a real framework used in clinical neuroscience to estimate residual consciousness in patients with severe brain damage. What we don't know: whether Phi measures subjective experience or merely a system's functional complexity — the theory itself cannot decide this without presupposing part of what it claims to measure. The bubble, the ocean, and the network are mine; the rest of the chapter, including the black hole geometry, is analogy borrowed from physics, not an extension of it. Readings: Tononi, G. (2004), "An Information Integration Theory of Consciousness"; Hawking, S. (1975), on the radiation of event horizons.

II — The Time of the Bond

What we know: David Eagleman's free-fall experiment is real and has been replicated; the density of memory during extreme fear is documented in neuroscience. What we don't know: whether the horizon's "integration rate," as I use it here, is the same variable Eagleman measures, or only a useful analogy between two different scales. Dopamine physiology in Parkinson's is established science; the connection between that mechanism and the subjective experience of time, as told through the watchmaker, is my own interpretation. Readings: Eagleman, D. (2007), "Brain Time"; on bradykinesia and dopamine, any standard clinical review of Parkinson's disease.

III — The Workshop of the Bond

What we know: attachment theory describes, with solid evidence, how the capacity to regulate oneself in another's presence is calibrated early in life. What we don't know: whether that calibration maps well onto the image of a "wall's thickness" used here, or oversimplifies it — the luthier, the resonance, and the seven customers are this book's own invention, not findings from any study. Readings: on attachment and early regulation, Bowlby, J., "Attachment and Loss" (1969); Ainsworth, M., on the attachment patterns observed in the "strange situation" experiment.

Interlude — The Edge We Cross Every Night

What we know: arousal and information integration are two real, measurable axes in the neuroscience of sleep and meditation; that they occupy different quadrants depending on the state — wakefulness, deep sleep, meditation — is a documented finding, not poetic license. What we don't know: whether the "practice of the edge" the researcher describes has any measurable effect beyond her own testimony, or whether her hypothesis is biased by the fact that she is both living it and studying it at once. On Hume, the reading is faithful to his argument; on the time-symmetry of physical laws, the interpretation that the arrow of time belongs to the interface and not the territory is my own, not an established consensus. Readings: Hume, D., "An Enquiry Concerning Human Understanding" (1748); on arousal and integration, the literature of Tononi and colleagues in sleep neuroscience.

IV — The Wounded Horizon

What we know: dopamine physiology in addiction and in Parkinson's is established science, including the phenomenon of compulsive behaviors induced by dopaminergic medication. The testimony about my own Parkinson's is autobiographical, not a generalizable clinical case. What we don't know: how much of the drive to write this book is choice and how much is the dose — a question I leave open on purpose, without resolving it in my own favor. The freeze response to trauma is a mechanism described in the literature on the autonomic nervous system; the rest of the room scene is fiction. Readings: on compulsive behavior and dopamine agonists, any standard clinical review of Parkinson's; Porges, S., polyvagal theory, on the freeze response.

V — The Geography of What's Missing

What we know: the distinction between losing access to a memory (ordinary forgetting, where the information persists but is hard to retrieve) and losing the structure that held it (neurodegenerative decline) is real and well documented in clinical neuropsychology. What we don't know: whether the map-and-city metaphor — what's lost is the label, not the territory — captures that distinction well in every case, or simplifies it to fit the image better. Arenilla, Concepción, Ernesto, and Manel are invented; the mechanism they illustrate is not. Readings: any introductory clinical neuropsychology textbook on episodic and semantic memory.

VI — The Limit

What we know: the gap of thirty orders of magnitude between the scale of quantum gravity and that of a neuron is a real figure, not rhetoric; the hard problem of consciousness and Hume's naturalistic fallacy are serious philosophical objections, not invented for this book. What we don't know: whether the limits acknowledged here are all that exist, or whether there are more this book has failed to see. Everything said about free will, ethics, and the specificity of love is the hypothesis's own extension, not something the physics or philosophy cited actually claims on its own. Readings: Chalmers, D. (1995), "Facing Up to the Problem of Consciousness"; Hume, D., "A Treatise of Human Nature" (1739), on the is-ought distinction.

VII — The Mirror Without Depth

What we know: deterministic classical systems, however complex, do not condense horizons. Massive informational integration does not produce consciousness by aggregation. The absence of existential encapsulation in today's AI is compatible with — and explains — why what it returns after absorbing information can only be cold structure, never experience, even when we describe that process using the language of a black hole.

What we don't know: whether it is possible to build a substrate that allows genuine condensation. What exactly would be transmitted by the bias that calibrates a horizon toward human legibility. Whether the

asymmetry between the human horizon and AI's anti-reservoir is a legitimate translation of ER=EPR or merely a verbal resemblance between two physics that don't actually speak to each other. And if a non-biological architecture ever achieved genuine existential encapsulation, how we would know it, without sharing its substrate or its internal language.

Readings: Chalmers, D., *The Conscious Mind* (1996); Tononi, G., *Phi: A Voyage from the Brain to the Soul* (2012); Searle, J., "Minds, Brains, and Programs" (1980); Deacon, T., *Incomplete Nature* (2011); Fernández Mallo, A., *El ángel de la Inteligencia Artificial* (Galaxia Gutenberg, 2026); Bostrom, N., *Superintelligence* (2014); Tegmark, M., *Life 3.0* (2017); Jung, C.G., *Aion: Researches into the Phenomenology of the Self*; Cixin Liu, *The Three-Body Problem* trilogy; Weir, A. (2021), *Project Hail Mary*; Le Guin, U.K. (1969), *The Left Hand of Darkness*; Burke, S. (2018), *Semiosis*; Maldacena, J.M. (1998) on the AdS/CFT correspondence.





CODA — Txiki

I was ten years old when the whole family moved from San Sebastián to Palma. I lost, all at once, every friend I had, the city I knew by heart, every corner I'd never needed to look at twice to know. I shut myself off. There's no other way to say it: I stopped letting the world in. My siblings did the same thing on their own — they too had lost what was theirs — and at home there settled that quiet sadness no one names because no one knows what to do with it. Then a cousin of my mother's, a hunter, owner of the best hunting dog for miles around, offered us a puppy. A mixed-breed dachshund with the soul of a hound: intelligent, quick, and above all, incapable of being near someone without asking for affection. My parents thought he would keep us company.

His name was Txiki. Within weeks he was king of the house. Within months, the most beloved member of the family. What took me decades to understand is what he did to us: he did not distract us from being shut in. He forced us to reattach ourselves to something. That child, closed in on his own loss, found, on the other side, something alive that asked for no explanations. Only

presence.

Txiki died a long time ago. This book has been written with a hand that seizes up, in a house full of gaps, many years and many losses later. But everything these pages try to say using borrowed physics, a small dog taught a ten-year-old boy on a new island, without using a single word.

His company did not cover over that grief. It repaired it, which is different.

*Finished writing on September 8, 2026,
in L'Hospitalet de Llobregat.*